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School of Engineering

Adaptation of Knowledge Integration in Innovation-Driven Organisations During Rapid Growth

A Case Study within Mechanical Development

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Abstract

Purpose – The purpose of this study was to develop a better understanding of how innovation-driven organisations adapt knowledge integration during rapid organisational growth.

Methodology – A qualitative single-case study was conducted within mechanical development in an innovation-driven organisation during rapid organisational growth. The case data were collected through a preliminary study, document studies, observations, semi-structured interviews, and informal interviews. The empirical material was analysed through thematic analysis.

Findings – The study identified three aggregate dimensions: Individualised and Informal Knowledge Integration, Formal Knowledge Mechanisms Misaligned with Work Practice, and Growth-Induced Fragmentation of Knowledge and Capabilities. The findings indicate that knowledge integration was largely carried out through tacit experience, informal interaction, personal networks, and individual responsibility. These practices supported flexibility and situated problem-solving but also created person-dependence and uneven access to knowledge. During rapid organisational growth these vulnerabilities became more visible as knowledge became increasingly distributed across employees, roles, functions, and projects. A key challenge was therefore not a lack of knowledge, but that knowledge became more difficult to locate, coordinate, and reuse.

Implications – The study contributes to knowledge integration literature by explaining how rapid organisational growth can intensify the tension between informal knowledge practices and formal knowledge mechanisms. The main practical implication is practice-aligned formalisation, meaning that organisations should strengthen formal mechanisms in ways that support, rather than replace, informal and situated knowledge integration. This includes making knowledge more visible, accessible, reusable, and connected to daily work.

Limitations – The study was conducted as a single-case study within one organisational context and focused on mechanical development and design-test-related work. The findings should therefore be understood as analytically generalisable to theory and potentially transferable to similar organisational contexts, rather than statistically generalisable. The proposed model of practice-aligned formalisation should also be understood as a practical implication derived from the analysis, not as an empirically tested solution.

Keywords – Knowledge integration; Rapid organisational growth; Innovation-driven organisations; Informal knowledge integration; Formal knowledge mechanisms; Tacit knowledge; Mechanical development; Qualitative single-case study

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1 Introduction

This chapter introduces the research area, research gap, and problem addressed in this study. It also presents the purpose, research questions, scope and delimitations, and outline of the thesis.

1.1 Background

Knowledge is a central resource in innovation-driven organisations, as innovation depends on the ability to create, combine, and apply knowledge across individuals, functions, and technical domains (Argote & Ingram, 2000; Kogut & Zander, 1992). In this thesis, innovation-driven organisations refer to technology-based organisations in which innovation is not merely a general ambition, but a key foundation for competitiveness, value creation, and technical product development (Vargas et al., 2020). In product development and problem-solving, relevant knowledge is often distributed across several actors rather than held by one individual or function (Argote & Ingram, 2000). Organisations therefore need to coordinate and combine specialised knowledge to make it useful in shared work processes (Argote et al., 2022; Grant, 1996).

The ability to combine specialised and distributed knowledge is commonly referred to as knowledge integration (Grant, 1996). Knowledge integration is important in innovation-driven organisations because innovation work often requires collaboration between individuals with different expertise, responsibilities, and perspectives (Lupton & Beamish, 2014; Watson et al., 2024). In such contexts, knowledge integration supports collective problem-solving, decision-making, and product development by making different knowledge contributions usable in shared work.

When organisations grow, the degree of specialisation between organisational units often increases, which creates a greater need for integration between differentiated functions and work processes (Lawrence & Lorsch, 1967). Different parts of the organisation may develop distinct skills, tasks, and areas of responsibility. This can increase the need to coordinate and combine knowledge across the organisation to support collaboration and the use of specialised expertise (Argote et al., 2022; Argote & Ingram, 2000).

The need for knowledge integration can become particularly challenging during rapid organisational growth. Under such conditions, organisational structures, roles, and work processes may change while daily work still needs to continue (Greiner, 1972). Organisational members may therefore face fragmentation, unclear interfaces, and coordination difficulties under time pressure, while collaboration between specialised functions remains necessary for innovation-driven work. How knowledge integration takes place during rapid organisational growth is therefore an important topic to examine.

1.2 Research Gap

Previous studies have conceptualised knowledge integration as a coordination problem that arises from the specialised and distributed nature of organisational knowledge (Grant, 1996; Okhuysen & Bechky, 2009). Research has also linked organisational growth to increasing specialisation, structural complexity, and coordination demands between organisational units (Greiner, 1972; Lawrence & Lorsch, 1967). These perspectives provide an important foundation for understanding why organisations need to integrate knowledge as they become larger and more specialised.

However, there remains limited empirical understanding of how knowledge integration is adapted in innovation-driven organisations during rapid organisational growth (Argote et al., 2022; Watson et al., 2024). This is particularly relevant because innovation-driven organisations often rely on collaboration between specialist functions to develop new solutions and manage complex tasks (Lupton & Beamish, 2014; Watson et al., 2024). When new roles, functions, and interfaces are introduced during growth, work practices may remain partly informal and not always clearly defined in formal organisational processes (Canonico et al., 2017).

Hence, there is a need for empirical studies that examine how knowledge integration is characterised in innovation-driven organisations and how rapid organisational growth affects the conditions for integrating knowledge. The study addresses that gap by focusing on how knowledge is accessed, shared, coordinated and applied in practice during rapid organisational growth.

1.3 Problem Description

For innovation-driven organisations, the practical challenge is not only to possess specialised knowledge, but to make this knowledge accessible and usable across organisational boundaries. Product development and problem-solving often involve several specialist functions, where employees need to access, interpret and combine knowledge from different parts of the organisation (Lupton & Beamish, 2014; Watson et al., 2024). If this integration is weak, knowledge may remain fragmented between individuals, roles, or functions, limiting the organisation's ability to use its expertise collectively (Argote & Ingram, 2000; Grant, 1996).

This challenge can become more difficult during rapid organisational growth. As organisations expand, the degree of specialisation between roles and units may increase, creating greater demands for integration between differentiated parts of the organisation (Lawrence & Lorsch, 1967). At the same time, structures, roles, and work processes may change as the organisation develops, which can make earlier ways of working less sufficient (Greiner, 1972). In such situations, employees may need to collaborate across a growing organisational structure without always having established ways of coordinating and applying relevant knowledge.

A further challenge is that knowledge integration can depend on both informal and formal mechanisms. Informal interaction and individual experience may be important because organisational knowledge is often tacit, situated, and difficult to fully codify (Nonaka, 1994). However, during rapid organisational growth, reliance on informal interaction may become difficult to sustain when more employees, roles and functions need to access knowledge. Formal mechanisms can support knowledge integration by clarifying how knowledge and work should be coordinated across the organisation (Mintzberg, 1979; Okhuysen & Bechky, 2009). However, such mechanisms need to be connected to actual work practices to support how knowledge is accessed, interpreted, and applied in daily work (Canonico et al., 2017).

The problem addressed in this study is therefore how innovation-driven organisations adapt knowledge integration when rapid organisational growth increases fragmentation, coordination demands, and the need for more scalable ways of sharing and applying knowledge.

1.4 Purpose and Research Questions

The purpose of this study is to develop a better understanding of how innovation-driven organisations adapt knowledge integration during rapid organisational growth.

To fulfil the purpose, two research questions have been formulated. The first research question addresses what characterises knowledge integration in innovation-driven organisations. It focuses on the practices, mechanisms, and conditions through which knowledge becomes accessible, shared, coordinated, and applied across roles and functions.

RQ1: What characterises knowledge integration in innovation-driven organisations?

The second research question addresses how rapid organisational growth affects knowledge integration. It focuses on how growth changes the conditions for knowledge accessibility, coordination, knowledge transfer, and the balance between informal and formal knowledge integration mechanisms.

RQ2: How does rapid organisational growth affect knowledge integration?

Together, the two research questions make it possible to analyse knowledge integration both as an organisational practice and as a practice affected by rapid organisational growth. This provides the basis for discussing how knowledge integration may need to be adapted during rapid organisational growth.

1.5 Scope and Delimitations

This study focuses on knowledge integration in innovation-driven organisations during rapid organisational growth. More specifically, the study addresses organisational and practical conditions for how knowledge is accessed, shared, coordinated, and applied across roles and functions. The study is limited to knowledge integration in product development processes, where specialised knowledge needs to be combined across functional boundaries.

The study does not focus on technical knowledge management systems, digital platforms, or IT-based solutions for storing and retrieving knowledge. It also does not evaluate product performance, financial effects, or strategic decision-making related to growth. The study is therefore limited to the organisational conditions surrounding knowledge integration and how these conditions are affected by rapid organisational growth.

1.6 Outline

Chapter 1 Introduction	*Introduces the research area and defines the problem addressed in the study. It also presents the purpose, research questions, scope and delimitations, and the outline of the thesis.*
Chapter 2 Theoretical framework	*Presents the theoretical basis for this study. The selected literature provides the concepts used to analyse knowledge integration in innovation-driven organisations during rapid organisational growth.*
Chapter 3 Methodology	*Describes the research design, case background, research process, data collection, and data analysis. It also addresses research quality, ethical considerations, and the use of artificial intelligence.*
Chapter 4 Empirical Findings	*Presents the empirical findings. The findings are structured around three aggregate dimensions derived from the empirical material.*
Chapter 5 Theoretical Analysis	*Analyses the empirical findings in relation to the theoretical framework and the two research questions. The chapter explains what characterises knowledge integration and how rapid organisational growth affects knowledge integration.*
Chapter 6 Discussion	*Discusses the findings in relation to the purpose of the study. It also presents theoretical and practical implications, limitations, suggestions for further research, and concludes the thesis.*

2 Theoretical framework

This chapter presents the theoretical lens used to analyse knowledge integration in innovation-driven organisations during rapid organisational growth. The theoretical framework focuses on knowledge integration as the core concept of the study, while also addressing formal and informal integration mechanisms, cross-functional collaboration, and the organisational consequences of rapid growth.

2.1 Connection between Research questions and Theory

The theoretical framework provides the conceptual foundation for analysing knowledge integration in innovation-driven organisations during rapid organisational growth. Since the study addresses what characterises knowledge integration and how rapid organisational growth affects knowledge integration, the framework is structured to support the analysis of both knowledge integration in practice and the organisational conditions that shape it.

Section 2.2 introduces knowledge in innovation-driven organisations and explains why knowledge integration is central when specialised and distributed knowledge must be combined in product development and problem-solving. Section 2.3 presents the distinction between tacit and explicit knowledge, which is used to analyse the relation between experience-based knowledge, documentation, and formal knowledge mechanisms. Section 2.4 defines knowledge integration as the core concept of the study and develops this concept through coordination and knowledge integration mechanisms. Section 2.5 positions cross-functional product development as an important context for knowledge integration, since innovation-driven work often requires knowledge to be coordinated across functional and disciplinary boundaries. Section 2.6 addresses rapid organisational growth and structural complexity, explaining how growth can increase differentiation, fragmentation, and coordination demands. Finally, Section 2.7 synthesises the theoretical framework and clarifies how different theoretical areas are used as an analytical lens for the empirical findings. Table 2.1 presents how the theoretical areas are used in connection to the research questions.

Table 2.1 Connection between research questions, theoretical areas, and analytical use

Research Questions	*Theoretical areas*	*Analytical use*
RQ1: *What characterises knowledge integration in innovation-driven organisations?*	*2.2 Knowledge in Innovation-driven Organisations; 2.3 Tacit and Explicit Knowledge; 2.4 Knowledge Integration; 2.4.1 Knowledge Integration as Coordination; 2.4.2 Knowledge Integration Mechanisms; 2.5 Knowledge Integration in Cross-functional Product Development*	*Used to analyse how knowledge is accessed, interpreted, coordinated, combined, and applied through experience, informal interaction, documentation, formal mechanisms, and cross-functional collaboration.*
RQ2*: How does rapid organisational growth affect knowledge integration?*	*2.5 Knowledge Integration in Cross-functional Product Development; 2.6 Rapid Organisational Growth and Structural Complexity; 2.7 Theoretical synthesis*	*Used to analyse how rapid growth changes the conditions for knowledge integration by increasing differentiation, fragmentation, newcomer learning needs, coordination demands, and dependence on formal and informal mechanisms.*

2.2 Knowledge in Innovation-driven Organisations

Knowledge is a central organisational resource in innovation-driven organisations because innovation depends on the ability to create, combine, and apply knowledge across individuals, functions, and technical domains. Organisational knowledge is not limited to individuals, but is embedded in members, technologies, procedures, tasks and the relationship between them (Argote & Ingram, 2000). This means that knowledge in organisations is distributed across both human and technical elements, which makes coordination and application central to how knowledge is used in practice (Argote et al., 2022; Argote & Ingram, 2000).

From a knowledge-based perspective, organisations can be understood as entities that coordinate and apply specialised knowledge. Grant (1996) argues that the primary role of the organisation is to integrate the specialised knowledge of individuals, while Kogut and Zander (1992) emphasise the organisation's ability to combine existing knowledge in productive ways. This is particularly relevant in innovation-driven organisations, where product development and problem-solving require knowledge from multiple specialised domains to be brought together in relation to shared tasks (Argote et al., 2022; Lupton & Beamish, 2014).

Innovation-driven organisations are characterised by a continuous need to generate and apply knowledge in response to technical, organisational, and market-related changes. Innovation activities often require collaboration across functional boundaries, since different domains contribute distinct expertise, perspectives and problem-solving approaches (Grant, 1996; Kogut

& Zander, 1992). When expertise is distributed across several actors and functions, organisations need mechanisms that support coordination, communication, and integration of knowledge across organisational boundaries (Argote et al., 2022; Manfredi Latilla et al., 2018).

Knowledge in innovation-driven organisations should not be understood as static information that can simply be stored and retrieved. Nonaka (1994) argues that organisational knowledge is created through interactions between individuals and through the relationship between tacit and explicit knowledge. This is important in innovation-driven work because much of the knowledge required for product development and problem-solving is experience-based, context-dependent, and distributed across different actors (Argote et al., 2022; Nonaka, 1994)

In this study, knowledge in innovation-driven organisations is understood as distributed, specialised, dynamic, and embedded in both individuals and organisational structures. This provides the basis for analysing why knowledge integration becomes central when specialised expertise must be accessed, interpreted, combined, and applied in collective work.

2.3 Tacit and Explicit Knowledge

The distinction between tacit and explicit knowledge is important for understanding how knowledge is shared, interpreted, and integrated in organisations. Nonaka (1994) conceptualises tacit knowledge as personal, context-specific, and closely connected to action, experience, values and practical engagement. Explicit knowledge, by contrast, refers to knowledge that can be articulated, codified, and communicated through formal and systematic language, such as documents, procedures, manuals, specifications, and databases (Nonaka, 1994; Nonaka et al., 1996). This distinction is relevant for knowledge integration because different forms of knowledge require different mechanisms to be shared, combined, and applied in organisational work (Kogut & Zander, 1992; Nonaka et al., 1996).

Tacit knowledge is closely connected to individual experience and practical know-how. Since tacit knowledge is developed through action and involvement in specific contexts, it is often difficult to separate from the individuals, tasks, and situations in which it is used (Nonaka, 1994; Nonaka et al., 1996). In organisational settings, tacit knowledge can be embedded in work practices, routines, problem-solving approaches, and professional judgement (Kogut & Zander, 1992; Lam, 2000). This makes tacit knowledge particularly important in innovation-driven organisations, where employees often rely on prior experience and contextual understanding when interpreting problems, making decisions, and applying specialised expertise (Argote et al., 2022; Nonaka, 1994).

Explicit knowledge differs from tacit knowledge because it can be expressed, documented, and transferred in more formalised ways (Nonaka, 1994). Explicit knowledge can support organisational coordination because it can be stored, communicated, and reused across individuals, functions and projects (Argote & Ingram, 2000; Grant, 1996). Documentation, process descriptions, templates, reports, and guidelines can therefore support knowledge integration by making knowledge available beyond the individual or situation in which it was

originally developed (Argote & Ingram, 2000). However, explicit knowledge only supports integration when it is accessible, understandable, and connected to actual work practice (Grant, 1996; Nonaka, 1994).

Tacit and explicit knowledge should not be understood as two isolated categories. Nonaka (1994) argues that organisational knowledge is created through interaction between tacit and explicit knowledge, which means that knowledge may move between experience-based practice and codified representations. This is relevant for knowledge integration because knowledge often needs to be articulated, documented, interpreted, and internalised before it can be reused in new tasks or by new organisational members (Argote et al., 2022; Nonaka, 1994). Effective knowledge integration therefore depends on the organisation's ability to support both interaction-based knowledge exchange and codified knowledge structures (Grant, 1996; Okhuysen & Bechky, 2009).

In innovation-driven organisations, the relationship between tacit and explicit knowledge is important because innovation depends on both experience-based judgement and formalised knowledge (Nonaka, 1994). Experience-based knowledge can support problem-solving, adaptation, and decision-making in uncertain situations (Lam, 2000; Nonaka, 1994), while explicit knowledge can support consistency, documentation, standardisation, and knowledge reuse across projects and functions (Argote & Ingram, 2000; Grant, 1996). If tacit knowledge remains strongly dependent on specific individuals, knowledge integration may become vulnerable when those individuals are unavailable (Lam, 2000). Conversely, if explicit knowledge is poorly structured, difficult to access, or disconnected from actual work practices, it may fail to support the coordination and reuse of knowledge (Argote & Ingram, 2000; Grant, 1996).

For this study, the distinction between tacit and explicit knowledge provides a basis for analysing both informal and formal aspects of knowledge integration. Tacit knowledge is used to analyse the role of accumulated experience, professional judgement, and situated problem-solving. Explicit knowledge is used to analyse why knowledge integration through documentation, process descriptions, templates, and other formal knowledge mechanisms. The distinction is therefore used to analyse why knowledge integration may become dependent on individual experience and informal interaction, while also explaining why formal mechanisms need to be accessible, structured, and aligned with actual work practices.

2.4 Knowledge Integration

Knowledge integration is the core concept of this study. It refers to the process through which specialised and distributed knowledge is combined and applied in collective organisational work. Grant (1996) argues that organisations exist not primarily to create knowledge, but to integrate the specialised knowledge held by individuals. Kogut and Zander (1992) similarly emphasise that organisations develop capabilities by combining existing knowledge in productive ways. From this perspective, knowledge integration concerns how different knowledge contributions are coordinated and used in practice, rather than merely how

knowledge is possessed by individuals or stored within the organisation (Grant, 1996; Kogut & Zander, 1992).

The need for knowledge integration arises because organisational knowledge is distributed across individuals, tools, tasks, routines, and relationships (Argote & Ingram, 2000). Argote & Ingram (2000) argue that knowledge is embedded not only in members of the organisation, but also in the interactions between members, tools, and tasks. This distributed nature of knowledge means that no single actor usually holds all the knowledge required to perform complex organisational work (Grant, 1996). Organisations therefore need to coordinate and combine knowledge across different actors, functions, and organisational contexts to support collective action (Argote et al., 2022; Grant, 1996).

Knowledge integration differs from knowledge transfer. Knowledge transfer concerns the movement of knowledge from one individual, group, or organisational unit to another (Argote & Ingram, 2000). Knowledge integration, by contrast, concerns how different forms of specialised knowledge are combined and applied in joint work (Grant, 1996). This distinction is important because access to knowledge does not automatically mean that knowledge is integrated (Argote et al., 2022; Argote & Ingram, 2000). For knowledge to support organisational outcomes, it must be interpreted, adapted, coordinated, and applied in relation to specific tasks and work practices (Argote et al., 2022; Grant, 1996).

Knowledge integration is particularly important in innovation-driven organisations because innovation often requires the combination of different knowledge domains (Kogut & Zander, 1992). Product development and technical problem-solving typically involve several specialised functions, each contributing different forms of expertise, assumptions, and priorities (Argote et al., 2022; Grant, 1996). In such contexts, knowledge integration enables organisations to transform distributed expertise into coordinated action (Grant, 1996). Without effective integration, knowledge may remain fragmented across individuals or functions, reducing the organisation's ability to use its specialised expertise collectively (Argote et al., 2022; Kogut & Zander, 1992).

For this study, knowledge integration is understood as the coordinated combination and application of specialised knowledge across individuals, functions, and organisational practices. This understanding makes it possible to analyse how knowledge is accessed, interpreted, coordinated, and applied in innovation-driven work. The concept is therefore used to analyse what characterises knowledge integration in the empirical findings and how rapid organisational growth affects the conditions under which knowledge can be integrated.

2.4.1 Knowledge Integration as Coordination

Building on the understanding of knowledge integration as the coordinated application of specialised knowledge, a coordination perspective clarifies how knowledge integration is achieved in practice. Coordination concerns the alignment of interdependent activities, tasks, and contributions to achieve collective outcomes (Okhuysen & Bechky, 2009). In relation to knowledge integration, this means that different knowledge contributions must not only be

available but also aligned with one another and applied in relation to shared organisational tasks (Grant, 1996; Okhuysen & Bechky, 2009).

Okhuysen and Bechky (2009) identify shared understanding, accountability, and predictability as important integrating conditions for coordination. Shared understanding enables actors to interpret how their own work relates to the work of others, while accountability clarifies responsibility for different contributions. Predictability allows actors to anticipate how others will act within shared work processes, which is important when several actors depend on each other's knowledge, decisions and actions (Okhuysen & Bechky, 2009).

Coordination can be supported through both formal and informal arrangements. Formal arrangements, such as routines, procedures, defined roles, and process descriptions, can guide how knowledge contributions are expected to be combined in recurring organisational activities (Grant, 1996; Xi et al., 2020). These arrangements can reduce uncertainty by providing structure for how work should be performed and how responsibilities are distributed (Grant, 1996; Okhuysen & Bechky, 2009). However, formal coordination may be insufficient when tasks are uncertain, knowledge is tacit, or work requires interpretation across functional boundaries (Carlile, 2004; Nonaka, 1994).

Informal coordination is therefore also important for knowledge integration. Interaction, dialogue, and direct communication enable actors to interpret context-dependent knowledge and adjust their actions in relation to others (Okhuysen & Bechky, 2009). This is particularly relevant when knowledge is experience-based or difficult to fully capture in formal documents and procedures (Nonaka, 1994). Informal coordination can therefore complement formal structures by enabling situated problem-solving and adaptation in practice (Okhuysen & Bechky, 2009; Xi et al., 2020).

The coordination of knowledge becomes more complex when knowledge crosses functional or disciplinary boundaries. Carlile (2004) argues that knowledge boundaries create challenges because actors from different functions may use different terminology, assumptions, priorities, and problem-solving approaches. In such situations, knowledge integration requires more than access to information, knowledge must also be translated and adapted so that it becomes meaningful and useful across functions (Carlile, 2004; Okhuysen & Bechky, 2009).

For this study, knowledge integration as coordination provides an analytical lens for examining how knowledge is aligned and applied in practice. This perspective supports the analysis of how employees rely on experience, informal interaction, documentation, processes, and cross-functional collaboration to coordinate knowledge. It also supports the analysis of how rapid organisational growth may increase coordination demand by expanding the number of actors, roles, functions, and interdependencies involved in shared work.

2.4.2 Knowledge Integration Mechanisms

Knowledge integration mechanisms refer to the organisational arrangements and practices that enable distributed knowledge to be coordinated, combined, and applied in collective work (Grant, 1996). Grant (1996) identified rules, directives, sequencing, routines, and group

problem-solving as mechanisms through which organisations integrate specialised knowledge. These mechanisms differ in how they support integration, as some rely on predefined structures while others depend on interaction between actors (Grant, 1996; Okhuysen & Bechky, 2009). This distinction is important because different types of knowledge require different mechanisms to be integrated effectively (Nonaka, 1994; Okhuysen & Bechky, 2009).

Formal mechanisms support knowledge integration by structuring how knowledge should be documented, accessed, transferred, and applied. Rules, procedures, routines, process descriptions, templates, and documentation can guide action and reduce the need for repeated problem-solving in recurring tasks (Grant, 1996). Documentation and formalised procedures can also help retain knowledge within the organisation by embedding it in tools, tasks, routines, and organisational artefacts rather than only in individuals (Argote & Ingram, 2000). In this way, formal mechanisms can support knowledge accessibility, consistency, and reuse across projects, functions, and organisational units (Argote & Ingram, 2000; Cummings & Teng, 2003).

However, formal mechanisms do not automatically lead to effective knowledge integration. For documentation, routines, and process descriptions to support integration, they need to be understandable, accessible, and connected to actual work practice (Argote & Ingram, 2000; Grant, 1996). If formal mechanisms are unclear, outdated, difficult to locate, or detached from how work is performed, they may provide limited support for the coordination and reuse of knowledge (Argote & Ingram, 2000; Cummings & Teng, 2003). This is particularly relevant when knowledge is tacit or context-dependent, since such knowledge cannot always be fully captured through formal documentation or procedures (Lam, 2000; Nonaka, 1994).

Informal mechanisms support knowledge integration through interaction, experience, and social relationships. When knowledge is difficult to articulate or when tasks involve uncertainty, employees often need to rely on dialogue, direct communication, and shared problem-solving to interpret and apply knowledge in practice (Nonaka, 1994; Okhuysen & Bechky, 2009). Informal networks can therefore help actors identify who possesses relevant knowledge and enable knowledge to be accessed through personal contact and situated interaction (Argote et al., 2022; Okhuysen & Bechky, 2009). Such mechanisms are important because they allow employees to adapt knowledge to specific contexts rather than only following predefined instructions (Grant, 1996; Nonaka, 1994).

Informal mechanisms can also compensate for limitations in formal structures, but they can also create vulnerabilities. When formal channels are slow, incomplete, or unclear, direct interaction can provide a faster way to access knowledge and solve problems (Okhuysen & Bechky, 2009). At the same time, reliance on informal mechanisms can make knowledge integration dependent on individual experience, personal networks, and knowing who to ask (Argote et al., 2022). This creates a potential vulnerability, as knowledge may remain unevenly distributed and difficult to access for newcomers or employees outside established networks (Argote & Ingram, 2000; Lam, 2000).

Boundary-spanning mechanisms are needed when knowledge must be integrated across functional, disciplinary, or organisational boundaries. Carlile (2004) argues that knowledge boundaries create challenges because actors from different functions may use different terminology, interpret information differently, and prioritise different aspects of a shared task. In such situations, knowledge integration requires more than the transfer of information, knowledge must also be translated and transformed so that it can be used across functional perspectives (Carlile, 2004). Boundary-spanning mechanisms, such as cross-functional meetings, design reviews, shared artefacts, and structured communication practices, can therefore support the interpretation and adaptation of knowledge across functions (Carlile, 2004; Cummings & Teng, 2003).

Effective knowledge integration depends on the combination of formal, informal, and boundary-spanning mechanisms. Formal mechanisms can support consistency, documentation, and reuse, while informal mechanisms support interpretation, adaptation, and experience-based problem-solving (Grant, 1996; Nonaka, 1994). Boundary-spanning mechanisms are important when knowledge needs to move across functions and be made meaningful in relation to different technical or organisational perspectives (Carlile, 2004). The effectiveness of knowledge integration therefore depends not only on whether mechanisms exist, but on how well they are combined and aligned with the knowledge being integrated and the work practices in which they are used (Grant, 1996; Okhuysen & Bechky, 2009).

For this study, knowledge integration mechanisms provide a basis for analysing how knowledge is integrated through both formal, informal, and boundary-spanning practices. Formal mechanisms are relevant for analysing documentation, process descriptions, and structured ways of working (Grant, 1996). Informal mechanisms are relevant for analysing experience-based knowledge, personal networks, direct interaction, and individualised knowledge seeking. Boundary-spanning mechanisms are relevant for analysing how knowledge is coordinated across functions in product development and how rapid organisational growth may increase the need for more deliberate integration practices (Argote et al., 2022; Carlile, 2004).

2.5 Knowledge Integration in Cross-functional Product Development

Cross-functional product development provides an important context for knowledge integration because product development activities often require the coordination of specialised knowledge from several functional domains. Cross-functional teams bring together individuals from different areas of expertise to support innovation and product development (Ancona & Caldwell, 1992). Since each function contributes different knowledge, assumptions, and priorities, cross-functional product development creates both opportunities for knowledge combination and challenges for knowledge coordination (Edmondson & Nembhard, 2009).

The need for cross-functional collaboration is closely connected to the distributed nature of organisational knowledge. Grant (1996) argues that organisations integrate specialised knowledge held by individuals, while Kogut and Zander (1992) emphasise the organisation's

capability to combine existing knowledge in productive ways. In product development, this means that technical, operational, and market-related knowledge must be brought together and applied in relation to shared product-related tasks (Ancona & Caldwell, 1992; Grant, 1996). Cross-functional collaboration can therefore be understood as a practical setting in which knowledge integration is enacted through interaction between specialised actors (Edmondson & Nembhard, 2009; Okhuysen & Bechky, 2009).

However, cross-functional collaboration does not automatically result in effective knowledge integration. Carlile (2004) argues that knowledge boundaries emerge when actors from different functions have different knowledge bases, terminology, interests, and interpretations of the same problem. These boundaries make it more difficult to transfer, interpret, and apply knowledge across functions (Carlile, 2004). As a result, cross-functional product development requires mechanisms that support not only the exchange of information, but also the translation and adaptation of knowledge across functional perspectives (Carlile, 2004; Okhuysen & Bechky, 2009).

Shared understanding is therefore central to cross-functional knowledge integration. Okhuysen and Bechky (2009) describe shared understanding as an important condition for coordination, as it enables actors to understand how their own work relates to the work of others. In product development, shared understanding helps actors align technical decisions, interpret requirements, and coordinate interdependent tasks across functional boundaries (Edmondson & Nembhard, 2009; Okhuysen & Bechky, 2009). Without shared understanding, different functions may interpret product requirements, design decisions, or process responsibilities differently, which can increase coordination difficulties (Carlile, 2004).

Cross-functional collaboration also depends on both formal and informal integration mechanisms. Formal mechanisms, such as design reviews, structured meetings, process descriptions, and shared documentation, can create arenas where knowledge from different functions is brought together (Cummings & Teng, 2003; Grant, 1996). Informal mechanisms, such as direct communication, personal networks, and everyday interaction, can support situated interpretation and problem-solving when knowledge is tacit, uncertain, or context-dependent (Nonaka, 1994; Okhuysen & Bechky, 2009). Effective cross-functional knowledge integration therefore depends on how formal and informal mechanisms are combined in practice (Grant, 1996; Okhuysen & Bechky, 2009).

In innovation-driven organisations, cross-functional product development is particularly important because innovation often involves uncertainty, technical complexity, and interdependence between specialised domains. Edmondson and Nembhard (2009) emphasise that innovation in teams depends on learning, collaboration, and the ability to work across disciplinary and functional boundaries. When knowledge is effectively integrated across functions, organisations can combine diverse expertise into more coherent product development decisions and innovation outcomes (Argote et al., 2022; Edmondson & Nembhard, 2009). When such integration is weak, knowledge may remain fragmented within functional areas,

reducing the organisation's ability to use its specialised expertise collectively (Carlile, 2004; Grant, 1996).

For this study, cross-functional product development is used as a theoretical context for analysing how knowledge integration is carried out across functions, roles, and areas of expertise. This perspective is relevant for examining how employees coordinate knowledge through formal meetings, informal interaction, documentation, processes, and shared product-related work (Carlile, 2004; Okhuysen & Bechky, 2009). It also supports the analysis of how rapid organisational growth may affect knowledge integration by increasing functional differentiation, reducing shared understanding, and making knowledge more fragmented across the organisation (Argote et al., 2022).

2.6 Rapid Organisational Growth and Structural Complexity

Rapid organisational growth changes the conditions under which knowledge integration takes place. Organisational growth involves more than an increase in size it also affects how work is divided, coordinated, and formalised within the organisation. Greiner (1972) argues that organisations face different managerial challenges as they grow, since structure and practices that were effective in earlier stages may become insufficient when the organisation becomes larger and more complex. This makes growth relevant for knowledge integration, as increasing scale can change both where knowledge is located and how it needs to be coordinated in practice (Argote et al., 2022; Greiner, 1972).

A central structural consequence of growth is increased division of work. Mintzberg (1979) describes organisational structure as the way work is divided into distinct tasks and how coordination is achieved between these tasks. As organisations expand, work often becomes more specialised, with clearer roles, functions, and areas of responsibility (Lawrence & Lorsch, 1967; Mintzberg, 1979). While specialisation can support the development of deeper functional expertise, it also increases the need for coordination mechanisms that reconnect differentiated activities (Lawrence & Lorsch, 1967; Mintzberg, 1979).

Differentiation is therefore central to understanding how rapid growth affects knowledge integration. Lawrence and Lorsch (1967) describe differentiation as the development of specialised units with different goals, time orientations, and ways of working. As differentiation increases, knowledge becomes more distributed across functions and organisational units, which increases the need for integration across those units (Lawrence & Lorsch, 1967). In knowledge-intensive organisations, this creates a structural challenge because specialised expertise must be coordinated and applied across organisational boundaries to support collective work (Argote et al., 2022; Grant, 1996).

Growth also increases structural complexity by expanding the number of actors, roles, functions, and interdependencies involved in organisational work. Mintzberg (1979) identifies several coordination mechanisms, including mutual adjustment, direct supervision, standardisation of work processes, standardisation of outputs, and standardisation of skills.

These mechanisms show that coordination can be achieved through both formalised structures and interaction between actors. In rapidly growing organisations, the challenge is that existing coordination mechanisms may no longer match the increased complexity of work (Greiner, 1972; Mintzberg, 1979).

This mismatch can create a lag between organisational growth and the development of appropriate structure, processes, and coordination practices. Greiner (1972) argues that growth creates new organisational problems when earlier ways of working no longer fit the organisation's increased size and complexity. Lawrence and Lorsch (1967) similarly emphasise that increased differentiation requires corresponding integration mechanisms. If integration mechanisms do not develop at the same pace as differentiation, organisations may experience unclear responsibilities, fragmented work practices, and difficulties coordinating across functions (Greiner, 1972; Lawrence & Lorsch, 1967).

This is particularly relevant for innovation-driven organisations, where knowledge integration depends on both formal and informal coordination. Formal mechanisms, such as defined roles, routines, process descriptions, documentation, and standardised procedures, can support coordination by clarifying how work should be carried out (Grant, 1996; Mintzberg, 1979). However, when organisations grow rapidly, formal mechanisms may become outdated, unclear, or insufficiently embedded in everyday work practices (Greiner, 1972; Watson et al., 2024). At the same time, informal coordination through personal networks, direct interaction, and mutual adjustment may become more difficult when the organisation becomes larger and more functionally divided (Mintzberg, 1979; Okhuysen & Bechky, 2009).

Rapid organisational growth can therefore contribute to fragmentation. Fragmentation occurs when knowledge, responsibilities, and work practices become distributed across specialised roles, functions, or organisational units without sufficient integration between them (Lawrence & Lorsch, 1967). Carlile (2004) argues that knowledge boundaries arise when actors from different functions use different terminology, assumptions, and priorities. When such boundaries are not actively managed, knowledge may remain within functions rather than being translated, adapted, and applied across shared work processes (Carlile, 2004; Lawrence & Lorsch, 1967).

In relation to knowledge integration, fragmentation is problematic because it can weaken shared understanding across the organisation. Okhuysen and Bechky (2009) identify shared understanding, accountability, and predictability as important conditions for coordination. When work becomes more specialised and distributed, employees may have detailed knowledge within their own function, but less understanding of how their work relates to the broader product, process, or organisational system. This can make it more difficult to coordinate specialised knowledge in collective work (Carlile, 2004; Okhuysen & Bechky, 2009).

Rapid growth also increases the importance of onboarding and knowledge sharing channels because more new employees need to gain access to organisational knowledge, experienced knowledge holders, and work practices. Onboarding can therefore be understood as part of the organisation's knowledge integration challenge during growth, rather than as a separate human

resource activity. New employees need to understand not only their own role, but also how their work relates to other actors, functions, processes, and knowledge sources.

From a knowledge integration perspective, onboarding supports new employees' access to explicit knowledge, such as procedures, expectations, and role-related information, as well as tacit knowledge, such as experience-based judgement and informal ways of working (Bauer et al., 2007; Klein et al., 2015; Nonaka, 1994). The relational dimension of onboarding is also important, since new employees need to identify who holds relevant expertise and how knowledge can be accessed through social relationships and organisational networks (Fang et al., 2011; Okhuysen & Bechky, 2009).

In rapidly growing organisations, weak onboarding can make knowledge integration more fragmented and person dependent. If onboarding does not provide sufficient role clarity, practical guidance, social access, and knowledge sharing channels, new employees may become dependent on individual initiative, informal questioning, and the availability of experienced employees (Bauer et al., 2007; Klein et al., 2015).

For this study, rapid organisational growth is understood as a contextual condition that increases structural complexity and changes the coordination requirements for knowledge integration. Growth can increase specialisation, differentiation, fragmentation, and newcomer learning needs, while also increasing the need for mechanisms that connect knowledge across roles, functions, and organisational units (Greiner, 1972; Lawrence & Lorsch, 1967; Mintzberg, 1979). This perspective supports the analysis of how rapid organisational growth affects knowledge integration by reshaping both formal structures, informal coordination practices, cross-functional collaboration, and onboarding (Argote et al., 2022; Okhuysen & Bechky, 2009).

2.7 Theoretical Synthesis

The theoretical framework positions knowledge integration as the coordinated combination and application of specialised and distributed knowledge in collective work. Grant (1996) argues that organisations integrate the specialised knowledge held by individuals, while Kogut and Zander (1992) emphasise the organisation's ability to combine existing knowledge in productive ways. Since knowledge is embedded in individuals, tools, tasks, routines, and relationships, knowledge integration requires mechanisms that enable knowledge to be accessed, coordinated, and applied across different parts of the organisation (Argote & Ingram, 2000). In innovation-driven organisations, this becomes particularly important because product development and innovation often require knowledge from multiple functional domains to be combined in relation to shared tasks (Argote et al., 2022; Grant, 1996).

The framework further positions knowledge integration as dependent on the interplay between informal, formal, and boundary-spanning mechanisms. Informal mechanisms, such as individual experience, direct interaction, personal networks, and shared problem-solving, are important because tacit and context-dependent knowledge often needs to be interpreted through

interaction (Nonaka, 1994; Okhuysen & Bechky, 2009). Formal mechanisms, such as documentation, routines, process descriptions, and defined roles, can support knowledge storage, accessibility, coordination, and reuse (Argote & Ingram, 2000; Grant, 1996). Boundary-spanning mechanisms are important when knowledge needs to be translated and adapted across functional perspectives in cross-functional product development (Carlile, 2004). Knowledge integration therefore depends not only on whether these mechanisms exist, but also on how well they support the knowledge being integrated and the work practices in which they are used.

The framework also explains why rapid organisational growth can change the conditions for knowledge integration. Greiner (1972) argues that structures and practices that work in earlier stages of organisational development may become insufficient as organisations grow. Growth can increase differentiation, specialisation, and structural complexity, meaning that knowledge becomes more distributed across roles, functions, and organisational units (Lawrence & Lorsch, 1967; Mintzberg, 1979). As differentiation increases, organisations need stronger integration mechanisms to reconnect specialised knowledge and maintain coordination across functions (Lawrence & Lorsch, 1967). If integration mechanisms do not develop at the same pace as growth, knowledge may become fragmented across functional or organisational boundaries.

Within the context of rapid organisational growth, onboarding becomes part of the broader knowledge integration challenge. As more employees enter the organisation, new employees need access to role clarity, practical guidance, social relationships, and knowledge sharing channels to participate in organisational work (Bauer et al., 2007; Klein et al., 2015). Onboarding can support access to explicit knowledge, such as procedures and role-related information, as well as tacit knowledge, such as experience-based judgement and informal ways of working (Klein et al., 2015; Nonaka, 1994). The relational dimension is also important, since new employees need to identify who holds relevant expertise and how knowledge can be accessed through organisational networks (Fang et al., 2011). Weak onboarding may therefore make knowledge integration more dependent on individual initiative, informal questioning, and access to experienced knowledge holders.

Taken together, the theoretical framework is used to analyse knowledge integration as a relationship between people, practices, and structures. For RQ1, the framework supports the analysis of how knowledge integration is characterised by tacit and explicit knowledge, informal coordination, formal mechanisms, and cross-functional collaboration. For RQ2, the framework supports the analysis of how rapid organisational growth affects knowledge integration by increasing differentiation, fragmentation, newcomer learning needs, and pressure on existing formal and informal mechanisms. This provides the analytical lens for examining how innovation-driven organisations adapt knowledge integration during rapid organisational growth. How such adaptation takes place in practice is therefore an important problem to examine empirically.

3 Methodology

This chapter presents the methodological approach used in the study. It describes the research design, case background, research process, data collection, data analysis, research quality, ethical considerations, and the use of artificial intelligence. The chapter aims to clarify how the study was conducted and how the methodological choices supported the purpose and research questions.

3.1 Research Design

This study adopts an exploratory qualitative single-case study design. The aim of the study was to develop a better understanding of how innovation-driven organisations adapt knowledge integration during rapid organisational growth. Since the study aimed to examine how knowledge integration is characterised and how rapid organisational growth affects knowledge integration, an exploratory design was considered appropriate. Exploratory research is suitable when the studied phenomenon is not fully understood and when the aim is to develop a deeper understanding of a problem area (Säfsten & Gustavsson, 2019).

A qualitative research approach was used because the research questions focus on what characterises knowledge integration and how rapid organisational growth affects it. Qualitative research is suitable when the aim is to understand experiences, interpretations, practices, and organisational processes in context (Patel & Davidson, 2019; Säfsten & Gustavsson, 2019). This approach was therefore appropriate for examining knowledge integration as a situated organisational phenomenon.

The study was conducted as a case study because knowledge integration was examined within its organisational context. Case studies are suitable when the aim is to investigate a phenomenon in depth and within its real-world context, especially when the boundary between the phenomenon and its context is not clear (Yin, 2018).

The study was designed as a single-case study. The case was selected because it provided an opportunity to study knowledge integration in an innovation-driven organisation during rapid organisational growth (Yin, 2018). The unit of analysis was knowledge integration within mechanical development. This made it possible to study how knowledge integration is carried out and affected within a specific organisational setting.

3.1.1 Case Background

The study was conducted within an innovation-driven organisation active in the development of advanced technical products for training and simulation. The organisation operates in a technically complex context where product development requires contributions from several areas of expertise. To maintain confidentiality, the organisation is described on a general level, and no company-specific names, products, or sensitive technical details are disclosed.

The organisation has developed from a smaller product-oriented company into a more integrated part of a larger corporate group. Historically, the organisation was characterised by technical development, product expertise, and entrepreneurial leadership. As the business expanded, the organisation gradually developed more structured ways of working, including project-oriented work and increased collaboration between different areas of expertise. A later organisational change integrated the company more closely into the corporate group, which changed its organisational position and strengthened the connection to broader corporate structures.

At the time of the study, the organisation was under rapid organisational growth. The growth had increased the number of employees, roles, functions, and organisational interfaces involved in development work. For example, one function in the organisation had tripled its number of employees in four years. The organisation was also structured around areas of expertise, which made it a relevant setting for studying knowledge integration across specialised roles and functions.

The empirical focus of the study was placed within mechanical development. This part of the organisation was selected because it involves collaboration between several roles and functions in development and testing-related activities. The work requires technical knowledge, product understanding, and experience from different parts of the organisation to be brought together in relation to shared development tasks.

More specifically, the study focused on knowledge integration in relation to design-test-related work within mechanical development. This area was considered relevant because it involves the translation of requirements, experience, and technical knowledge into development-related decisions and test activities. The case therefore provided an opportunity to study how knowledge integration takes place within an innovation-driven organisation during rapid organisational growth.

3.1.2 Research Process

The research process in this study was carried out in several stages, each contributing to the development of the study's focus and empirical exploration. The process began with an initial literature review and a preliminary study, which were conducted partly in parallel. While the literature review supported the theoretical understanding of the research area, the preliminary study provided an initial understanding of the organisational context. Together, these activities informed the formulation of the problem statement, purpose, and research questions. The study then proceeded to the main data collection, analysis, and report writing. Although the process is described as a sequence of stages, the work involved continuous movement between theoretical understanding, empirical insight, and analytical development.

The initial literature review focused on the fields of knowledge management, knowledge transfer, knowledge integration, cross-functional collaboration, and organisational growth. The aim was to establish a theoretical understanding of the research area and to identify relevant

concepts and previous research. This phase helped clarify the broader problem area and provided a foundation for understanding knowledge integration as the central concept.

In parallel, a preliminary study was carried out to develop a basic understanding of the organisation's working methods and empirical context. The preliminary study included formal meetings, participation in workshops, initial observations, and informal interviews with employees in the organisation. Through these activities, an initial understanding was gained of how knowledge-related activities were organised in practice and which organisational areas were relevant to examine further. This phase supported the definition of the empirical focus of the study.

Based on insights gained from the literature review and the preliminary study, the study's problem statement, purpose and research questions were formulated. This step was important because the focus of the study developed through the combination of theoretical understanding and empirical orientation. The research questions were therefore formulated after the preliminary study, when the empirical context and the relevance of knowledge integration during rapid organisational growth had become clearer.

Subsequently, the main data collection was carried out. This consisted of semi-structured interviews, document studies, observations, and supplemented by informal interviews. These methods were used to gather empirical material on how knowledge was accessed, shared, coordinated, and integrated within the organisational context being studied. The different data sources contributed complementary perspectives on the phenomenon and supported a broader understanding of knowledge integration in practice.

The empirical material was then analysed through thematic analysis. In this stage, the collected material was reviewed, coded, and organised into themes and aggregated dimensions. The thematic analysis is described in more detail in Section 3.3. The final stage consisted of writing and refining the report. During this stage, the empirical results were separated from the theoretical analysis to clarify the distinction between what was identified in the empirical material and how these findings were interpreted through the theoretical framework.

3.2 Data Collection

The empirical data were collected through multiple qualitative data sources: semi-structured interviews, document studies and observations. These data sources were complemented by a preliminary study and informal interviews, which were used as supplementary sources of contextual understanding. The preliminary study supported the definition of the empirical focus, while informal interviews helped clarify organisational terminology, work practices, and questions that emerged during the study. Together, the data sources provided insight into how knowledge integration was described, documented, and discussed in relation to daily work practices. Table 3.1 provides an overview of the data sources and their use in the study.

Table 3.1 Overview of empirical data sources and their use in the study

Data source	Scope	Documentation	Use in the study
Preliminary study	Formal meetings, participation in two workshops, initial observations, and informal interviews	Field notes	Supported contextual understanding, helped define the empirical focus, and informed the formulation of problem description, purpose and research questions
Document studies	Internal webpages, process descriptions, role descriptions, workshop and meeting material in form of PowerPoints	Document notes and extracted relevant content	Provided insight into how structures, roles, processes, and work practices were formally described
Observations	3 observations: One task force meeting, two workshops, 60-150 minutes each	Field notes	Provided insights into how knowledge-related issues, responsibilities, and process-related questions were discussed in organisational settings
Semi-structured interviews	8 interviews, 40-60 minutes each	Audio recording and transcripts	Provided insight into employees' experiences, interpretations and descriptions of knowledge integration in practice
	Respondent, role, and duration:	*Respondent A: Senior Mechanical Engineer: 40 minutes* *Respondent B: Production Leader: 60 minutes* *Respondent C: Application Engineer: 60 minutes* *Respondent D: Mechanical Engineer: 50 minutes* *Respondent E: Senior Mechanical Engineer: 60 minutes* *Respondent F: Quality & Testing: 50 minutes* *Respondent G: Project Manager: 40 minutes* *Respondent H: Mechanical Engineer: 60 minutes*	
Informal interviews	Informal conversations with employees during the study	Field notes	Supported contextual understanding and helped clarify questions that came up during the study

3.2.1 Preliminary Study

A preliminary study was conducted to gain an initial understanding of the organisational context and to clarify the empirical focus of the study. In exploratory studies, an initial orientation phase can support the researcher in becoming familiar with the field and developing a more informed understanding of the phenomenon before the main data collection is carried out (Patel & Davidson, 2019; Säfsten & Gustavsson, 2019). In this study, the preliminary study was used to understand the organisation's working methods, development-related activities, and knowledge-related challenges.

The preliminary study consisted of formal meetings, participation in two workshops, initial observations, and informal interviews with employees in the organisation. These activities provided an overview of how work was organised and where knowledge integration appeared relevant in practice. During this phase, field notes were taken to document observations, reflections, and issues that appeared relevant to the study.

The preliminary study had two main purposes. First, it supported the development of contextual understanding, which was important because knowledge integration was studied within its organisational setting. Second, it helped define the empirical focus of the study by identifying relevant activities, roles, and organisational areas for further investigation. The preliminary study therefore contributed to the formulation of the problem description, purpose, and research questions, as well as to the planning of the main data collection.

The preliminary study was not used as the main empirical basis for the results. Instead, it functioned as an orientation phase that supported the design of the study and the interpretation of the empirical context.

3.2.2 Interviews

Semi-structured interviews were used as one of the empirical data sources in the study. Interviews are suitable in qualitative research when the aim is to gain a deeper understanding of individuals' experiences, perceptions and interpretations of practices and situations in their work (Brinkmann & Kvale, 2018). In case study research, interviews are also commonly used to gather insights into organisational processes and the perspectives of actors involved in the studied phenomenon (Yin, 2018).

The interviews were conducted in a semi-structured format. An interview guide was used to ensure that key themes were addressed in all interviews, while still allowing flexibility in the order and wording of the questions. This format enabled respondents to elaborate on their experiences and perspectives, while also supporting comparability across the interviews (Brinkmann & Kvale, 2018; Rowley, 2012). The interview guide was adapted slightly depending on the respondent's role and relation to the studied work. The interview guide is presented in Appendix A.

Before the main interviews were conducted, the interview guide was tested through a pilot test. The purpose of the pilot test was to assess whether the questions were clearly formulated and

relevant to the organisational context. Based on the pilot test, minor adjustments were made to the wording of some questions, and one additional question was added to the interview guide.

The selection of respondents was based on their roles, experience, and involvement in the process being studied. A purposive sampling approach was used to identify individuals with relevant knowledge of the studied phenomenon (Rowley, 2012). During the interview process, respondents were also asked to identify other individuals with insight into the process. This helped identify further relevant participants and supported an understanding of which roles and functions were involved in the work being studied.

A total of eight interviews were conducted, each lasting between approximately 40-60 minutes. The interviews were conducted individually and recorded with the participants' consent to ensure accurate documentation of the empirical data. The interviews were then transcribed and used as part of the thematic analysis. To protect the anonymity of the participants, respondents are referred to as Respondent A-H throughout the report.

3.2.3 Document Studies

Document studies were used as one of the empirical data sources in the study. In case study research, documents can provide insights into the organisational context and can be used to understand how processes, structures, and responsibilities are formally described (Yin, 2018). In this study, document studies were used to examine how the organisation formally represented work processes, roles, responsibilities, and knowledge-related practices.

The documents included internal webpages, process descriptions, role descriptions, organisational information, and workshop and meeting material in the form of PowerPoint presentations. Most documents were accessed through the organisation's internal platform, while some materials were provided by employees involved in the studied activities. The documents were selected based on their relevance to mechanical development, development-related work processes, and the organisational areas connected to the empirical focus of the study.

The document studies served two main purposes. First, they provided an understanding of how the organisation formally described structures, roles, and processes relevant to the studied work. Second, they made it possible to compare formal descriptions of work with how employees described and discussed knowledge integration in interviews and observations. This was important because the study aimed to understand knowledge integration both as formally represented and as carried out in practice.

During the document studies, notes were taken on content related to work processes, responsibilities, role descriptions, available knowledge resources, and process-related guidance. Relevant content was extracted and used as part of the empirical material in the thematic analysis. The document studies therefore contributed to understanding how formal knowledge mechanisms were represented within the organisation and how these mechanisms related to the studied phenomenon.

3.2.4 Observations

Observations were used as one of the empirical data sources in the study. Observations are suitable in case study research because they allow the researcher to study activities, interactions, and discussions as they occur within their real-life context (Yin, 2018). In this study, observations were used to gain insight into how knowledge-related issues, responsibilities, and process-related questions were discussed in organisational settings.

Three observations were conducted: one task force meeting and two workshops. The researcher was present during these activities and primarily took the role of non-participant observer. This meant that the focus was on observing discussion, interactions, and behaviours as they occurred, rather than actively influencing the situation. The observations provided insight into how employees discussed work processes, responsibilities, knowledge needs, and coordination across roles and functions.

Field notes were taken during and directly after the observations. The notes focused on aspects relevant to the study's purpose and research questions, including how participants discussed knowledge, how responsibilities were clarified, how process-related uncertainties were raised, and how different perspectives were brought together in the discussions. These notes were later included in the thematic analysis together with the interview material and document studies.

3.2.5 Informal interviews

Informal interviews were used as a supplementary source of contextual understanding during the study. As the researcher was present within the organisation during the research process, informal conversations with employees occurred naturally in relation to daily work, meetings, workshops, and follow-up questions. Such informal interviews can provide valuable contextual insight and help the researcher to understand meanings, practices, and situations that may not fully emerge in formal interviews (Swain & King, 2022).

In this study, informal interviews were not used as a main empirical data source. Instead, they were used to clarify questions that came up during the study, support the researcher's understanding of the organisational context, and provide additional background to the interpretation of the empirical material.

Notes from these interactions were documented in the form of field notes. These notes focused on contextual information, clarifications, and reflections relevant to the study's purpose and research questions. The informal interviews were not treated as separate interview data but were used to support the interpretation of patterns identified in the main empirical data sources.

3.3 Data Analysis

The empirical material was analysed through thematic analysis. The aim of the analysis was to identify recurring patterns related to knowledge integration and rapid organisational growth across the collected material. The analysis included interview transcripts, documents, observation notes, and field notes.

The analysis was conducted iteratively, meaning that the empirical material was revisited several times throughout the analytical process. The research questions guided the analysis focusing attention on how knowledge integration was characterised and how rapid organisational growth affected the conditions for knowledge integration. At the same time, the analysis remained open to patterns in the empirical material that developed during coding.

3.3.1 Thematic Analysis

Thematic analysis was used to analyse the empirical material, following the approach outlined by Braun and Clarke (2006). Thematic analysis is a method for identifying, analysing, and reporting patterns, or themes, within qualitative data (Braun & Clarke, 2006). It was considered suitable for this study because it provides a flexible approach to qualitative analysis and can be applied to different types of qualitative material.

In this study, thematic analysis was applied to interview transcripts, documents, observation notes, and field notes. The material was analysed as part of one coherent analytical process, where data from different sources were coded and compared to identify recurring patterns related to knowledge integration and rapid organisational growth.

The analysis followed the main phases of thematic analysis, including familiarisation with the data, coding, development of themes, reviewing themes, defining and naming themes, and producing the report. The process is illustrated in Figure 3.1. Although the phases are presented sequentially, the analysis was conducted iteratively, meaning that the material, codes, themes, and research questions were revisited throughout the process.

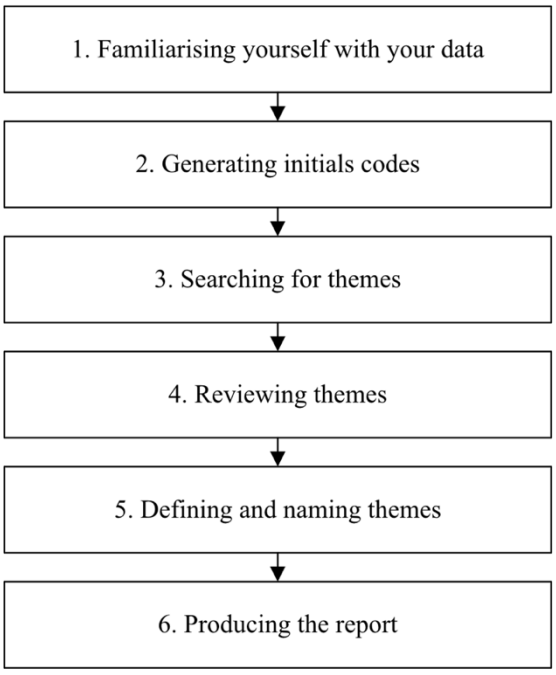


Figure 3.1 Overview of the thematic analysis process

Comment. The figure illustrates the main phases of the thematic analysis process, based on Braun and Clarke (2006). In this study, the phases were applied iteratively rather than a strictly linear sequence.

The analysis began with data familiarisation with the empirical material. Interview recordings were transcribed, and transcripts, documents, observation notes, and field notes were read several times to gain an overall understanding of the material. During this stage, notes were made regarding recurring issues, similarities, differences, and aspects that appeared relevant to the research questions.

The next stage involved coding the empirical material. Meaningful segments of text were identified and assigned codes that captured their central meaning. The coding focused on how knowledge was accessed, shared, coordinated, documented, interpreted, and applied, as well as how rapid organisational growth affected these activities. Codes were developed close to the empirical material and were continuously reviewed and refined as the analysis progressed.

After the initial coding, codes with similar meaning were compared and grouped into preliminary themes. These themes were then reviewed in relation to both the coded extracts and the full dataset to ensure that they represented coherent and relevant patterns. Some themes were merged, while others were revised or excluded when they did not sufficiently contribute to answering the research questions.

The final themes were developed into aggregate dimensions that captured the main empirical patterns in the material. These dimensions formed the basis for the presentation of empirical findings in Chapter 4. The themes were later interpreted in relation to the theoretical framework in Chapter 5, where empirical findings were analysed in relation to the two research questions.

3.3.2 Analytical Data Structure

To increase transparency in the analytical process, the results of the thematic analysis were organised into an analytical data structure inspired by Gioia et al. (2013). The data structure was used as a complementary way of presenting the analytical process, rather than as a separate analysis method. The purpose was to make visible how the empirical material was transformed from detailed observations and statements into broader themes and aggregate dimensions.

The first-order concepts were developed close to the empirical material and reflected recurring expressions, descriptions, and observations identified across the interviews, document studies, observations, and field notes. These first-order concepts were then compared and grouped into second-order themes, which represented more abstract interpretations of the empirical patterns. Finally, the second-order themes were synthesised into aggregate dimensions that captured the

main patterns explaining knowledge integration in the studied organisation. This is visualised in Figure 3.2.

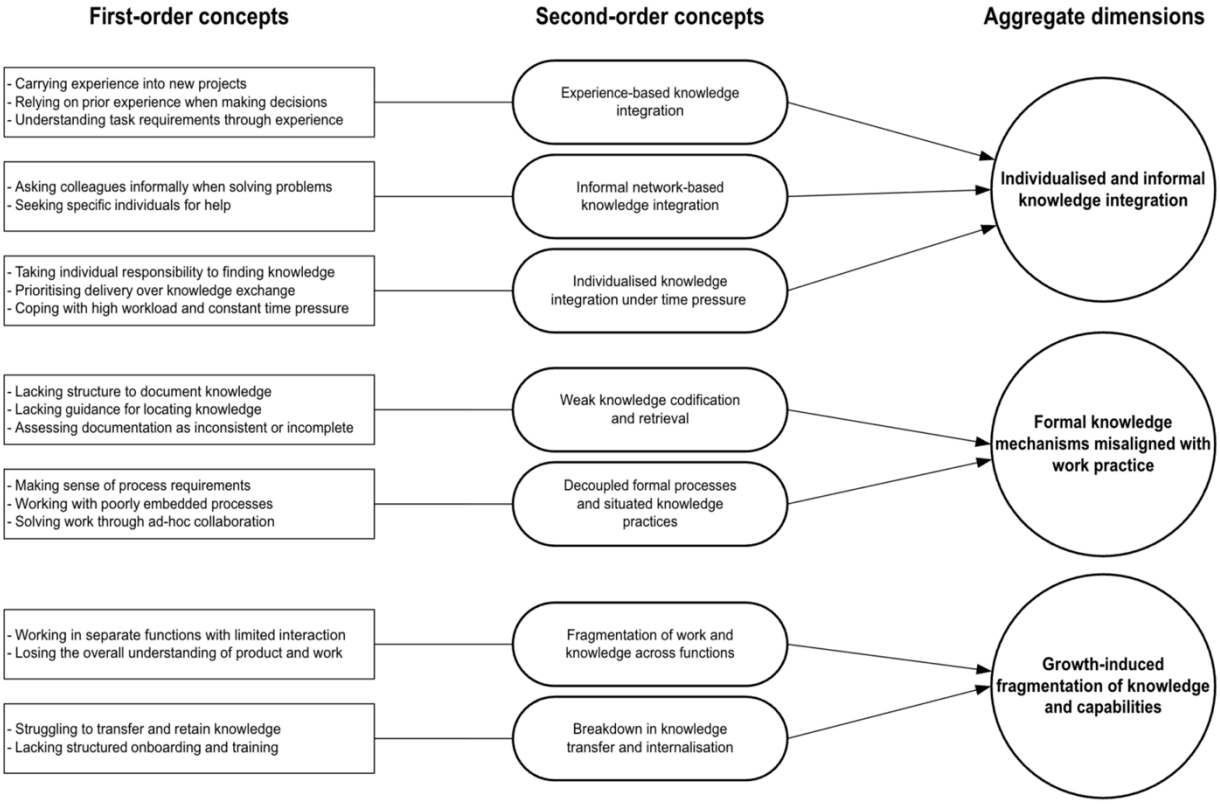


Figure 3.2 Gioia-Inspired analytical data structure

Comment. The figure illustrates how first-order concepts were grouped into second-order themes and synthesised into aggregate dimensions.

This structure supported the transparency and traceability of the analysis by clarifying the link between empirical material, codes, themes, and aggregate dimensions. The final analytical structure consisted of three aggregate dimensions: individualised and informal knowledge integration, formal knowledge mechanisms misaligned with work practice, and growth-induced fragmentation of knowledge and capabilities. These dimensions formed the basis for the empirical findings presented in Chapter 4 and the theoretical analysis developed in Chapter 5.

The full data structure, including aggregate dimensions, second-order themes, first-order concepts, and supporting empirical evidence, is presented in Appendix C.

3.4 Research Quality

In qualitative research, scientific quality can be discussed through the concept of trustworthiness. Trustworthiness includes credibility, transferability, dependability, and confirmability, which together address whether the study is credible, contextually transferable, transparent and grounded in the empirical material (Lincoln & Guba, 1985; Shenton, 2004).

Table 3.2 summarises how these quality criteria were addressed in the study. The table provides an overview of the measures taken.

Table 3.2 Overview of research quality criteria and measures taken in the study

Quality Criterion	Addressed in the study	Contribution to Quality
Credibility	Multiple data sources were used, including semi-structured interviews, document studies, observations, informal interviews, and a preliminary study. Interpretations were also discussed with academic supervisor.	Reduced dependence on a single data source and supported a more grounded understanding of the studied phenomenon.
Transferability	The organisational context, case background, empirical focus, and respondent roles were described.	Enabled reader to assess whether the findings may be relevant to similar organisational contexts.
Dependability	The research process was documented through the research design, data collection, data analysis, interview guide, pilot test, transcripts, coding, and analytical data structure.	Increased transparency and made the research process easier to follow.
Confirmability	The findings were grounded in empirical material through triangulation, documentation of the analysis, and reflection on the researcher's role.	Strengthened the link between empirical material and findings and reduced the influence of researcher bias.

The following section explains how each criterion was addressed in relation to design, data collection, data analysis, and interpretation of the study.

3.4.1 Credibility

Credibility concerns whether the findings provide a believable and well-founded account of the studied phenomenon (Lincoln & Guba, 1985; Shenton, 2004). In this study, credibility was strengthened through the researcher's continuous engagement with the organisation during the research process, including the preliminary study, observations, and informal interaction with employees. This supported contextual understanding and reduced the risk of interpreting the empirical material without sufficient knowledge of the setting.

Credibility was also strengthened using multiple data sources. The study combined semi-structured interviews, document studies and observations. This enabled comparison between what respondents described in interviews, how work was formally represented in documents,

and how knowledge-related issues were discussed in observed organisational settings. Such triangulation can strengthen qualitative research by reducing dependence on a single source of evidence (Shenton, 2004; Yin, 2018).

The preliminary study also contributed to credibility by helping the researcher understand the organisational context before the main data collection was carried out. This supported the formulation of a relevant empirical focus and helped ensure that the selected respondents, documents, and observations were connected to the purpose and research questions of the study.

In addition, interpretations were discussed continuously with the academic supervisor and, when relevant with the organisational representatives. These discussions were used to challenge emerging interpretations and reduce the risk of drawing conclusions that were not grounded in the empirical material.

3.4.2 Transferability

Transferability concerns the extent to which the findings may be relevant in other contexts (Lincoln & Guba, 1985; Shenton, 2004). In qualitative research, transferability is not achieved through statistical generalisation, but through providing sufficient contextual detail so that readers can assess whether the findings may be applicable to similar settings (Lincoln & Guba, 1985).

In this study, transferability was supported by describing the organisational context, the case background, the empirical focus, and the roles of the interview respondents. The study was conducted in an innovation-driven organisation during rapid organisational growth, with a focus on knowledge integration within mechanical development. By describing this context, the study enables readers to assess whether the findings may be relevant to other innovation-driven, knowledge intensive, or rapidly growing organisations.

Transferability was also supported through the selection of respondents with different roles and experience related to the studied work. This provided several perspectives on knowledge integration and helped capture variation within the case.

3.4.3 Dependability

Dependability concerns the transparency and the consistency of the research process (Lincoln & Guba, 1985; Shenton, 2004). In qualitative research, this means that the research process should be sufficiently documented so that the reader can understand how the study was conducted and how the findings were developed.

In this study, dependability was supported through a structured description of the research design, research process, data collection, and data analysis. The study describes how the initial literature review and preliminary study informed the formulation of the problem description, purpose, and research questions, and how the empirical material was collected through interviews, document studies, observations, and informal interviews.

Dependability was further supported using an interview guide, a pilot test, field notes, recorded interviews, and interview transcriptions. These measures contributed to a more systematic data collection process. The analysis was also documented through coding, theme development, and an analytical data structure inspired by Gioia et al. (2013). This increased transparency by making the movement from empirical material to first-order concepts, second-order themes, and aggregate dimensions more visible.

3.4.4 Confirmability

Confirmability concerns the extent to which the findings are grounded in the empirical material rather than shaped by researcher bias (Lincoln & Guba, 1985; Shenton, 2004). Since qualitative research involves interpretation, confirmability requires transparency regarding how interpretations were developed and how the researcher's role may have influenced the study.

In this study, confirmability was strengthened through triangulation of multiple data sources. By comparing interview material with documents, observations, and field notes, the analysis did not rely only on one type of empirical material. This supported the grounding of interpretations in patterns across the dataset rather than in isolated statements.

Confirmability was also supported through documentation of the analytical process. Interview transcripts, field notes, document notes, codes, themes, and aggregate dimensions were used to create traceability between the empirical material and the findings. The Gioia-inspired data structure further supported this by making the relationship between empirical observations and analytical interpretations more explicit.

The researcher's role was also considered during the study. Since the researcher was present within the organisation, there was a need to remain aware of how closeness to the context could influence interpretation. This was addressed through continuous reflection, documentation of analytical decisions, and discussion with academic and organisational supervisors. These measures helped ensure that the findings remained grounded in the empirical material and connected to the purpose and research questions of the study.

3.5 Ethical Considerations

Ethical considerations were addressed throughout the research process, particularly in relation to data collection, informed consent, confidentiality, and the handling of empirical material. In qualitative research, the researcher is responsible for protecting participants' integrity, privacy and rights during the study (Miles et al., 2014).

Before the interviews were conducted, all participants received information about the purpose of the study, how the empirical material would be used, and how their participation would be handled. Participants were informed that participation was voluntary and that they could withdraw from the study at any time. Informed consent was obtained before the interviews were recorded, in line with ethical principles for research involving participants (Säfsten &

Gustavsson, 2019). The information letter and consent form used in the study are presented in Appendix B.

To protect participants' anonymity, names and other directly identifying details have been excluded from the report. Respondents are instead referred to as Respondent A-H. The roles of respondents are described at a general level to provide transparency while reducing the risk of identifying individual participants. Interview recordings, transcripts, field notes, and document notes have been handled only within the scope of the study.

Since the study was conducted within an organisation, particular attention was also given to the handling of organisation-specific information. Company-specific names, sensitive technical details, and confidential internal information have been excluded or described at a general level. This was done to protect both individual participants and the organisation while still allowing the study to present relevant empirical findings.

3.6 Use of Artificial Intelligence

Generative artificial intelligence (AI) was used as support during the writing and revision process of this thesis. The tool used was ChatGPT by OpenAI. AI was used to support language refinement, text structuring, improvement of coherence between different parts of the report, and understanding of selected scientific articles.

The use of AI was limited to writing support, feedback, revision, and support in understanding selected literature. AI was not used to conduct the empirical data analysis. It was also not used to generate codes, define themes, interpret empirical material, or draw conclusions. The thematic analysis, interpretation of findings, analytical synthesis, and conclusions were developed and critically assessed by the author.

AI-generated suggestions were used as support in the writing process and were not inserted directly into the thesis. The final wording, structure, analysis, interpretations, and conclusions were developed by the author. No directly identifying participant information, interview transcripts, or sensitive company-specific information were shared with the AI tool. The author remains fully responsible for the content, analysis, interpretations, conclusions, and final wording of the thesis.

4 Results

This chapter presents the empirical findings of the study. The findings are structured around three aggregate dimensions and related to second-order themes developed from the empirical material. Selected quotations and empirical examples are included to illustrate the main patterns, while additional supporting empirical material is presented in Appendix C. The purpose of the chapter is to report the empirical patterns identified in the collected material, while theoretical interpretation is addressed in Chapter 5.

4.1 Individualised and Informal Knowledge Integration

The first aggregate dimension concerns individualised and informal knowledge integration. Across the empirical material, employees accessed, interpreted, and combined knowledge through accumulated experience, informal networks, and personal responsibility. This aggregate dimension consists of three second-order themes: experience-based knowledge integration, informal network-based knowledge integration, and individualised knowledge integration under time pressure.

4.1.1 Experience-based knowledge integration

Accumulated individual experience played a central role in how employees accessed and used knowledge across projects and work tasks. Employees described experience as something that was carried into new projects and used to understand what tends to work, what tended to fail, and what types of tests or design considerations are needed. One respondent explained:

> *"Partly, you know what holds up and what breaks, that's experience you carry with you, so you also know, more or less, what kinds of tests you need to run."* (Interview: Respondent A)

Another respondent similarly stated:

> *"Experience is something we need to carry with us somehow and bring into a project."* (Interview: Respondent C)

These statements illustrate how the employees described experience as important for judging what actions, tests, and design considerations were relevant in specific situations. Experience was also described as important when employees needed to interpret current problems in relation to previous projects and practical situations.

The observations reflected a similar pattern. During one workshop, participants described that part of the process existed within individuals and that this had become more challenging with more new employees in the organisation. A similar issue was raised during a task force meeting, that parts of the process existed within individuals rather than being clearly documented.

The empirical material also contained examples of experience being used when making decisions under uncertainty. Employees described situations where they relied on previous experience to judge what should be tested, what could be accepted, or how a specific design issue should be handled. For example, one respondent explained that decisions were partly based on experience and partly on reasoning through the available material:

"Partly, you rely on your experience, and partly, you think things through when you have the materials in front of you." (Interview: Respondent A)

Experience was also described as necessary for understanding task requirements. Respondents explained that certain tasks, such as designing or writing specifications, required an understanding that was difficult to acquire only through formal documents. One respondent stated:

"This is the kind of experience you'd say they need in order to be able to design." (Interview: Respondent C)

Taken together, the empirical material presents experienced employees as important carriers of knowledge. Their knowledge was not only presented in documents or process descriptions, but is embedded in their ability to interpret situations, recognise patterns, and connect tasks to previous experience.

4.1.2 Informal Network-Based Knowledge Integration

The second-order theme concerns the role of informal networks in employees' access to knowledge. Employees frequently described how they asked colleagues informally when problems occurred, rather than relying solely on formal meetings, systems, or documentation. One respondent explained:

"These aren't official meetings, but rather informal chats where you come down to talk, and maybe you've run into a problem, and then someone can say, 'Well, you could do it like this." (Interview: Respondent B)

Another respondent described a similar practice:

"You just go to colleagues sitting nearby who you thought might be able to help." (Interview: Respondent D)

These informal interactions enabled employees to access knowledge quickly and to solve problems that required input from others. The interview material also included several examples of employees relying on specific individuals who were known to possess relevant expertise. One respondent described how personal networks shaped knowledge access:

"Personal networks and it's from the site. They call me directly because they know me, and then they get my answer." (Interview: Respondent C)

Another respondent explained:

"Then you can write as many documents and processes as you like, but if I need someone to do something on the electronics side, I already know who to go to." (Interview: Respondent E)

Across these examples, informal networks appeared as practical routes for accessing knowledge. They helped employees identify where expertise was located, access practical knowledge, and coordinate across tasks and functions.

However, the empirical material also pointed to uneven access to knowledge through informal networks. Employees with established networks could access knowledge more easily, while newer employees could struggle to identify who possessed relevant expertise. This was visible in an observation where a participant asked:

"I'm new and don't really know who's good at what. Who among you are the experienced ones? Who of you has the knowledge?" (Observation 2: Workshop)

The observation illustrates how access to relevant expertise could be more difficult for employees who were new to the organisation or who did not yet know where specific knowledge was located.

4.1.3 Individualised Knowledge Integration under Time Pressure

The third second-order theme concerns individual responsibility for finding, interpreting, and integrating knowledge. Respondents described that in practice, they often had to reach out to others, ask for input, and determine whose knowledge was relevant. One respondent stated:

"In practice, it's more your own responsibility to reach out, ask, and get their input." (Interview: Respondent H)

The same respondent described how this could lead to inconsistent knowledge access:

"Depending on who you ask, you get different answers." (Interview: Respondent H)

The respondents further described that knowledge-seeking could vary depending on who was nearby or considered relevant at the time, rather than following an established approach. This describes knowledge access as dependent on individual initiative rather than a shared or systematic approach. Employees often needed to search for knowledge themselves and decide

which knowledge sources to rely on. This was described as particularly challenging when employees were new, when expertise was spread, or when formal forums for knowledge exchange were unclear.

The empirical material also contained several examples of time pressure and workload limiting opportunities for knowledge exchange. Several respondents described how project delivery, customer demands, and testing deadlines reduced the time available for knowledge sharing, verification, and support. One respondent explained:

> *"Something we maybe didn't have, since we weren't that many before... was proper knowledge sharing and exchange. And we also didn't have the time for it, because we had to focus on projects all the time."* (Interview: Respondent F)

Another respondent stated:

> *"We've effectively had 110% workload... and we simply haven't had the time to support development, neither in terms of knowledge transfer nor testing."* (Interview: Respondent C)

The empirical material therefore points to limited opportunities for knowledge sharing under conditions of time pressure and high workload. Employees described how they relied on their own initiative, immediate contacts, and available experience when formal opportunities for knowledge exchange were limited.

4.2 Formal Knowledge Mechanisms Misaligned with Work Practice

The second aggregate dimension concerns formal knowledge mechanisms and their alignment with daily work practices. Documentation, internal webpages, process descriptions, templates and formal routines are available in the studied organisation, but they are not always structured, retrievable, complete, updated, or embedded in actual work practices. This aggregate dimension consists of two second-order themes: weak knowledge codification and retrieval and decoupled formal processes and situated knowledge practices.

4.2.1 Weak Knowledge Codification and Retrieval

The empirical material contained several examples of knowledge being partly codified in documents, reports, internal webpages, templates, and process descriptions. However, employees expressed uncertainty about how documentation should be structured and where relevant information could be found. One respondent stated:

> *"I haven't found any general structure for how such a report should be written; it's more based on your own judgement."* (Interview: Respondent H)

Another respondent questioned whether existing reports are used as a source of knowledge:

"I would be very surprised if anyone goes in and reads old reports, it's probably quite heavy going." (Interview: Respondent B)

These statements point to a distinction between the existence of documentation and employees' ability to use documentation as a practical source of knowledge. Even when knowledge had been documented, employees did not always describe it as accessible, understandable, or useful for future work.

The document study supported this pattern. Several internal webpages contained information about working methods and other work-related content, but the structure and level of detail varied between pages. Some webpages included information about the latest update, while most pages did not. One webpage was explicitly marked as out of date, and some pages contained only limited information. Other parts of the internal website consisted mainly of clusters of links without a clear structure, and some links did not lead to any accessible content.

The empirical material points to documentation as available but unevenly structured and difficult to use consistently. Employees also expressed uncertainty about where information was stored. During the observations, questions such as "Where is the documentation stored?" and "Where do I find templates for this?" were raised. These questions illustrate how formal knowledge mechanisms did not consistently provide a clear route to relevant knowledge.

Taken together, these findings indicate that formal documentation is available, but that weak structure, limited searchability, inconsistent updating, and unclear ownership reduce its usefulness in daily work. As a result, employees often rely on individuals or informal networks to locate knowledge that could otherwise have been supported by formal knowledge mechanisms.

4.2.2 Decoupled Formal Processes and Situated Knowledge Practice

The empirical material also pointed to a gap between formal processes and daily work practices. Respondents and workshop participants described uncertainty regarding process requirements, responsibilities, terminology, and expectations. One respondent explained:

"You can see that the phase exists, but not really what should be done." (Interview: Respondent G)

This statement illustrates how the formal process provides a general structure but does not always clarify what actions are expected in practice. Workshop observations reflected similar patterns. Participants discussed that the process was difficult to understand, that different terms were interpreted in different ways, and that responsibilities were not always clear. The document study also identified variation in available content across process phases, with some phases described in more detail and others only presented as headings.

In one workshop, participants explicitly questioned why the process phases were not explained and why several abbreviations were used without clarification:

"Why are there not explanations for each phase and what is supposed to be done? And why is there a lot of abbreviations? Is there any document or something to explain these abbreviations?" (Observation 3: Workshop)

This illustrates how the process required interpretation before it could be used as practical support in daily work.

The formal process was also described as insufficiently introduced and inconsistently used. One respondent stated:

"I've never been part of any project so far that has followed the process completely." (Interview: Respondent H)

During a workshop, participants also stated that the process had been created without sufficient inclusion of employees and that they had not been working according to it. These examples describe a formal process that exists as a representation of how work should be performed, but that it is not fully integrated into how work is carried out in practice.

Instead, employees described how work was often coordinated through ad-hoc collaboration. They contacted different functions, asked for tests, arranged design reviews, or solved issues through informal and situation-specific interactions. One respondent described having several contact points across the organisation and explained that the interface varied depending on what needed to be solved. These descriptions suggest that employees adapt their knowledge-seeking behaviour to the situation, rather than following one clearly established process.

The observation also reflected how the process was still being interpreted and negotiated by employees. During workshops, participants discussed what different process steps should mean, who should be involved, and how the process should be used in practice. In this sense, the formal process appeared not only as a tool for coordinating work, but also an object of ongoing clarification and interpretation.

Taken together, the empirical material points to a gap between formal knowledge mechanisms and situated work practices. Formal processes, documentation, and templates exist, but they do not fully guide how knowledge is exchanged, interpreted, or coordinated in practice. Employees therefore rely on ad-hoc collaboration, informal communication, and individual judgement to bridge the gap between formal structures and actual work needs.

4.3 Growth-Induced Fragmentation of Knowledge and Capabilities

The third aggregate dimension concerns growth-induced fragmentation of knowledge and capabilities. The empirical material describes how rapid organisational growth has been associated with more specialised, distributed, and functionally separated work. At the same time knowledge transfer, onboarding, and capability development have not developed at the same pace as the organisation's increasing complexity. This aggregate dimension consists of

two second-order themes: fragmentation of work and knowledge across functions and breakdown in knowledge transfer and internalisation.

4.3.1 Fragmentation of Work and Knowledge Across Functions

The empirical material describes work and knowledge as increasingly divided across functions. Respondents described how the organisation previously worked in a more co-located and product-centred way, while the current organisation is more divided by professional functions. One respondent explained:

> *"Now you have mechanical engineers on one floor, doing their own thing, living in their own world. Then you have the electronics people doing the same, and software doing their own thing as well. We're not working together around the product. We're organising ourselves around our professions instead."* (Interview: Respondent C)

Another respondent described the change as follows:

> *"There weren't that many of us back then, and everyone basically did everything. Now we're more divided, so organisationally, we work differently."* (Interview: Respondent A)

These statements describe a change from broader roles and closer proximity towards a more specialised, and functionally divided organisation. When the organisation was smaller, fewer employees were involved, roles were broader and proximity between functions supported informal coordination. As the organisation has grown, work has become more specialised and distributed, which respondents described as making knowledge more difficult to access and combine across functions.

Employees also described a loss of overall product and process understanding. One respondent explained how broader project involvement had gradually been replaced by narrower work packages:

> *"Previously, we were more involved in the project as a whole... Now, it's more about work packages being handed out... and as a result, you lose the overall picture."* (Interview: Respondent A)

Respondents connected narrower work packages with a reduced understanding of how specific task related to the broader product, process, or organisational context. Instead of participating in the whole development process, employees become responsible for more limited parts of the work. This was described as creating a greater dependence on coordination across specialised roles and tasks.

The observation material also supported this pattern. During workshops related to the mechanical development process, participants discussed uncertainties regarding who should be

involved in different steps, how knowledge should move between functions, and how responsibilities should be distributed. These discussions illustrate that fragmentation was not only experienced at an individual level, but also visible in collective attempts to clarify the process.

Taken together, the empirical material describes rapid organisational growth as associated with a more functionally divided organisation. Knowledge is distributed across different roles and departments, while employees describe reduced proximity, reduced whole-product understanding, and increased dependence on coordination across functional boundaries.

4.3.2 Breakdown in Knowledge Transfer and Internalisation

The second second-order theme concerns knowledge transfer, onboarding, and capability development. The organisation has recruited many new employees, but the empirical material points to insufficient structured knowledge transfer and training. One respondent described this as:

> *"Neglected newcomers too many new hires, not enough training."* (Interview: Respondent C)

Another respondent explained that new employees had largely been expected to learn by themselves:

> *"Lately, it's felt like there's been a lot of new things coming in, and a lot of new people as well, and that they've mostly just had to jump in, learn on their own, and take individual responsibility."* (Interview: Respondent H)

These statements connect onboarding and capability development to individual responsibility and informal knowledge access. New employees often needed to ask senior colleagues, search for information, or learn through practice, rather than being supported by a systematic onboarding structure. This created a situation where knowledge transfer depended heavily on individual initiative and the availability of experienced employees.

One respondent described how this affected both newcomers and experienced employees:

> *"So, the easiest option for him is to ask me about everything, whenever he comes across something he doesn't know how to handle. And that puts quite a bit of pressure on me. So, it's really this onboarding phase that's a bit challenging new people may find it difficult to know where to find things."* (Interview: Respondent H)

Respondents also described knowledge retention as a challenge when knowledge is held by individuals and when certain products, tests, or tasks are performed infrequently. Some knowledge was described as difficult to maintain because it is used irregularly, while other knowledge was described as vulnerable when experienced employees leave or retire. These

examples point to a dependence on individuals when important knowledge is not systematically transferred, documented, or internalised by others.

The observations further illustrated how newcomers could struggle to identify where relevant knowledge is located. In one workshop, a participant explicitly asked who possessed experience and knowledge in the area. These examples point to a lack of visibility regarding where expertise is located within the organisation, not only a lack of formal training.

Taken together, the empirical material describes a capability gap connected to rapid organisational growth. The organisation has increased in size and complexity, but mechanisms for transferring, internalising, and retaining knowledge have not developed at the same pace. As a result, knowledge integration becomes dependent on senior employees, informal mentoring, and individual learning. This places pressure on experienced employees and makes knowledge integration more difficult for newcomers.

4.4 Summary of Empirical Findings

Taken together, the empirical material describes knowledge integration in the studied organisation as strongly dependent on people, relationships, and situated work practices. Employees appeared as important carriers of knowledge, while informal contacts helped employees locate expertise and solve problems in daily work. Knowledge integration was therefore not only connected to formal access to information but also to employees' ability to know who to ask, how to interpret previous experience, and how to apply knowledge in specific work situations.

At the same time, formal knowledge mechanisms provided uneven support. Documentation, internal webpages, process descriptions, templates, and formal routines are available, but the empirical material described them as difficult to locate, inconsistently structured, unevenly updated, and not fully embedded in daily work practices. As a result, employees often relied on informal communication, ad-hoc collaboration, and personal judgement to bridge the gap between formal structures and practical work needs.

The empirical material also describes how rapid organisational growth had changed the conditions for knowledge integration. Work had become more specialised, distributed, and functionally divided. Employees described reduced proximity, weaker product understanding, and increased dependence on coordination across roles and functions. At the same time, knowledge transfer, onboarding, and capability development had not developed at the same pace as the organisation's increasing complexity. This made knowledge integration more dependent on senior employees, informal mentoring, and individual learning.

Overall, the empirical findings point to a central empirical pattern, knowledge integration was maintained through informal and individualised practices, while formal mechanisms provided only partial support. Rapid organisational growth made this pattern more visible by increasing the need to access, transfer, and coordinate knowledge across a larger and more specialised organisation. The empirical findings summarised in this chapter are further supported by the empirical material presented in Appendix C. These empirical findings form the basis for the analysis in chapter 5.

5 Analysis

In this chapter the results are analysed through the theoretical lens of existing literature and in relation to the two research questions.

5.1 RQ1: What characterises knowledge integration in innovation-driven organisations?

This section analyses the empirical findings through the theoretical framework to address the first research question. From the theoretical framework, knowledge integration can be understood as the coordinated combination and application of specialised and distributed knowledge in collective work (Argote et al., 2022; Grant, 1996; Kogut & Zander, 1992). This means that knowledge integration is not only about whether knowledge exists within the organisation, or whether it has been transferred from one person, function, or project to another. For knowledge to become integrated, it must be interpreted, connected to other forms of knowledge, and applied in relation to specific organisational tasks and work practices (Argote & Ingram, 2000; Grant, 1996; Okhuysen & Bechky, 2009).

In the studied organisation, knowledge integration is characterised by three connected patterns. First, knowledge integration is strongly dependent on tacit and experience-based knowledge. Second, it takes place through informal and relational coordination between employees. Third, formal knowledge mechanisms exist but they only partially support how knowledge is integrated in practice. Together, these patterns indicate that knowledge integration in the studied innovation-driven organisation is active, situated, and dependent on the relationship between people, work practices, and formal structures.

5.1.1 Knowledge integration as tacit, experience-based practice

The first characteristic of knowledge integration in the studied organisation is its dependence on tacit and experience-based knowledge. The empirical findings indicate that experienced employees play a central role in interpreting technical problems, identifying relevant tests, judging what is likely to work, and connecting current development to previous projects. Knowledge integration therefore depends not only on access to information, but also on the practical judgement required to understand what information means in a specific context.

This can be understood through Nonaka's (1994) distinction between tacit and explicit knowledge. Tacit knowledge is personal, context-specific, and closely connected to action and experience, and practical involvement (Nonaka, 1994). In the studied organisation, much of the knowledge required for development work appears to have this character. Employees do not only need documented information about previous projects, tests, or processes. They also need the ability to interpret that information, relate it to the current situation, and decide how it should influence action.

This becomes important because innovation-driven work often involves uncertainty and technical complexity. In such situations, knowledge cannot always be fully separated from the

people and situations in which it has been developed. Lam (2000) argues that tacit knowledge is often embedded in work practices, routines, and professional judgement. This helps explain why experience becomes so central in the empirical findings. Experienced employees function as carriers of organisational knowledge because they can recognise patterns, anticipate problems, and connect previous experience to current tasks.

The findings therefore nuance Grant's (1996) view of the organisation as a mechanism for integrating specialised knowledge. The studied organisation does not only need to bring specialised knowledge together. It also needs to make experience-based knowledge interpretable and applicable across tasks, projects, and functions. Specialised knowledge becomes organisationally useful when it is connected to collective work, not only when it is held by individuals. This interpretation is also consistent with Kogut and Zander's (1992) argument that organisation develops capabilities by combining knowledge in productive ways.

However, the empirical findings also show that the dependence on tacit knowledge creates vulnerability. When knowledge is embedded in experienced employees' judgment, it becomes difficult to access for others unless those employees are available, involved, or known to be relevant knowledge holders. This creates a form of person-dependence. Tacit knowledge supports flexible problem-solving, but it also makes knowledge integration fragile when knowledge remains concentrated in specific individuals rather than being supported by accessible structure, routines, or shared practices (Argote & Ingram, 2000; Lam, 2000; Nonaka, 1994)

The implication is that tacit knowledge integration in the studied organisation is characterised by a strong reliance on experience as an integrating resource. This is important in an innovation-driven context because product development and technical problem-solving often require judgement that cannot be fully codified. At the same time, this reliance means that knowledge integration becomes dependent on individuals' ability to remember, interpret, and apply previous experience. Thus, tacit and experience-based knowledge is both a necessary condition for knowledge integration and a source of vulnerability when it is not made accessible beyond experienced employees.

5.1.2 Knowledge integration as informal and relational coordination

The second characteristic of knowledge integration in the studied organisation is its dependence on informal and relational coordination. The empirical findings indicate that employees frequently accessed knowledge through direct contact, informal conversations, and personal networks. This pattern can be interpreted through the coordination perspective presented by Okhuysen and Bechky (2009). They argue that coordination depends on shared understanding, accountability, and predictability. In the studied organisation, informal interaction appears to support these conditions in practice. Through direct conversations, employees clarify problems, identify dependencies, and develop a shared understanding of what needs to be done. Informal coordination therefore becomes a practical mechanism for integrating knowledge, especially when tasks are uncertain, knowledge is tacit, or formal processes provide limited guidance (Nonaka, 1994; Okhuysen & Bechky, 2009).

The analytical point is that informal networks are not only social relationships, in this case they have an organisational function. They help employees locate expertise, interpret unclear situations, and combine knowledge across roles and functions. This is consistent with Argote and Ingram's (2000) view that organisational knowledge is embedded not only in individuals, but also in tools, tasks, and the relationships between them. In the studied organisation, knowledge is therefore embedded both in formal artefacts and in the social relations that make it possible to access and use knowledge in daily work.

This is particularly relevant in cross-functional development work. Carlile (2004) argues that knowledge boundaries emerge when actors from different functions have different knowledge bases, terminology, interests, and interpretations. In the studied organisation, informal coordination helps employees move across such boundaries by enabling direct explanation, clarification, and translation of knowledge. This means that informal interaction does not only support the exchange of information, but it also supports the interpretation of knowledge across functional perspectives (Carlile, 2004; Okhuysen & Bechky, 2009).

However, the relational character of knowledge integration also creates uneven access to knowledge. Employees with established networks can access knowledge quickly because they know who to ask. New employees or employees outside of established networks, may not have the same ability to identify relevant knowledge holders. The issue is therefore not only whether knowledge exists within the organisation but whether employees know how to find it, and whether they have access to the relationships through which knowledge becomes available.

The analytical implication is that informal coordination is necessary but not sufficient. It is necessary because tacit and experience-based knowledge often requires interaction to be interpreted and applied. It is not sufficient because informal networks do not guarantee equal, visible, or scalable access to knowledge. In relation to RQ1, knowledge integration in the studied organisation is therefore characterised by relationally embedded coordination, where informal interaction makes knowledge usable in practice but also makes knowledge access dependent on personal networks and individual initiative.

5.1.3 Formal mechanisms as partial support for knowledge integration

The third characteristic of knowledge integration in the studied organisation is that formal mechanisms exist but provide only partial support for knowledge integration in practice. The organisation has documentation, internal webpages, process descriptions, templates, and formal routines. These mechanisms can be understood as attempts to codify knowledge and make it accessible beyond individual employees and specific situations.

From a theoretical perspective, such mechanisms are important for knowledge integration. Grant (1996) identifies rules, directives, routines, sequencing, and group problem-solving as mechanisms through which organisations integrate specialised knowledge. Similarly, Argote and Ingram's (2000) argue that knowledge can become embedded in organisational tools, tasks, and routines. In principle, formal mechanisms should therefore reduce dependence on

individual knowledge holders and support knowledge reuse across projects, functions, and organisational units (Argote & Ingram, 2000; Cummings & Teng, 2003).

However, the empirical findings indicate that the existence of formal mechanisms does not automatically lead to knowledge integration. Documentation and process descriptions only support integration when employees can find them, understand them, and apply them in relation to actual work. In the studied organisation, formal mechanisms are often perceived as difficult to locate, unevenly updated, unclear, or insufficiently connected to daily work practices. This means that knowledge may exist in formal forms without becoming practically usable knowledge.

This can be interpreted through the relationship between explicit and tacit knowledge. Nonaka (1994) describes explicit knowledge as knowledge that can be codified and communicated through formal language. However, explicit knowledge still requires interpretation. A process description, report, or template does not integrate knowledge by itself. Employees need to understand how it relates to their task, who is responsible for using it, and how it should guide action. When formal mechanisms lack this connection to situated work, employees continue to rely on informal interaction and individual judgement.

This is also consistent with Cummings and Teng's (2003) argument that knowledge transfer is affected by the characteristics of the knowledge, the source, the recipient, and the context in which transfer takes place. In the studied organisation, the problem is not only that knowledge needs to be documented. The problem is also that documented knowledge needs to be retrievable, understandable, and connected to the work situation where it is supposed to be used. If this connection is weak, documentation becomes a storage mechanism rather than an integration mechanism (Argote & Ingram, 2000; Cummings & Teng, 2003).

The findings therefore point to a misalignment between formal knowledge mechanisms and situated knowledge practices. Formal processes may indicate that a phase exists but still provide limited guidance regarding what should be done, who should be involved, or how knowledge moves between functions. Using Okhuysen and Bechky (2009) terminology, unclear processes and responsibilities weaken shared understanding, accountability, and predictability. As a result, formal mechanisms provide structure at a general level but insufficient support for knowledge integration in practice.

This interpretation does not mean that formal mechanisms are irrelevant. Rather, it suggests that their value depends on how well they are embedded in work practice. Documentation, templates, and process descriptions can support knowledge integration when they help employees retrieve, interpret, and reuse knowledge in relation to actual tasks (Cummings & Teng, 2003; Grant, 1996). When they are detached from practice, they risk becoming formal artefacts rather than active integration mechanisms.

The analytical implication is that formal mechanisms in the studied organisation are present but not fully embedded. They exist as organisational artefacts, yet they are not always sufficiently integrated into how employees search for, interpret, and apply knowledge. In relation to RQ1,

knowledge integration is therefore characterised by a dual pattern: formal mechanisms do exist, but informal and experience-based practices remain central for making knowledge usable in daily work.

5.1.4 Summary of answer to RQ1

The analysis of RQ1 highlights that knowledge integration in the studied innovation-driven organisation is characterised by the interaction between tacit experience, informal coordination, and partially supportive formal mechanisms. Knowledge integration is not a linear process where knowledge is transferred from one place to another. Rather, it takes place when employees interpret, adapt, and combine knowledge in relation to specific development tasks.

This means that knowledge integration is strongly practice-based. Employees need to understand not only where knowledge is located, but also how it should be used in a specific situation. Experienced employees therefore become important carriers of organisational knowledge because they can connect previous projects, technical judgement, and practical experience to current problems.

At the same time, knowledge integration is relational. Employees often access knowledge through informal conversations and personal networks. This makes knowledge integration flexible and useful in daily work, especially when formal documentation or processes do not provide enough guidance. However, it also means that knowledge access becomes uneven. Employees who know whom to ask can integrate knowledge more easily than newcomers or employees outside established networks.

Formal mechanisms are therefore important, but in the studied organisation they only provide partial support. Documentation, internal webpages, process descriptions, templates, and routines exist, but they are not always sufficiently accessible, updated or connected to how work is performed. Overall, RQ1 is answered by showing that knowledge integration is characterised by tension between situated informal practice and formal mechanisms that have not yet become fully embedded in daily work.

5.2 RQ2: How does rapid organisational growth affect knowledge integration?

This section analyses the empirical findings through the theoretical framework to address the second research question: How does rapid organisational growth affect knowledge integration? While Section 5.1 focused on what characterises knowledge integration in the studied organisation, this section analyses how rapid organisational growth changes the conditions under which specialised and distributed knowledge can be accessed, coordinated, and applied.

The analysis indicates that rapid organisational growth affects knowledge integration by increasing both the need for integration and the difficulty of achieving it. In the studied organisation, growth has increased the number of employees, roles, functions, projects, and interdependencies involved in development work. As a result, knowledge has become more

distributed across the organisation, while the need to connect and combine this knowledge has become more demanding (Argote et al., 2022; Lawrence & Lorsch, 1967).

A central interpretation is that rapid organisational growth does not only increase the amount of knowledge within the organisation. It also changes where knowledge is located, how it is accessed, and what mechanisms are required to make it usable across functions. Previous forms of knowledge integration based on proximity, broad roles, personal networks, and informal interaction appear to have become less sufficient as the organisation has expanded.

Interpreted through the theoretical framework, rapid organisational growth can therefore be understood as creating a stronger need for knowledge integration while simultaneously making existing integration practices more difficult to maintain. Growth increases differentiation, specialisation, and structural complexity, which makes knowledge more fragmented across roles and functions. At the same time, informal and person-dependent mechanisms become more difficult to rely on when more employees need access to knowledge and when work becomes more functionally divided (Argote & Ingram, 2000; Okhuysen & Bechky, 2009).

The following sections analyse how rapid organisational growth affects knowledge integration by increasing differentiation and coordination demand, fragmenting knowledge and capabilities, and exposing the limits of existing knowledge integration mechanisms.

5.2.1 Rapid growth increases differentiation and coordination demand

The first way in which rapid organisational growth affects knowledge integration is by increasing differentiation and coordination demand. In the studied organisation, growth has been associated with a shift from broader more product-centred roles towards a more specialised and functionally divided organisation. Work that was previously carried out with closer proximity between functions has become more distributed across professional areas, roles, and work packages.

This can be interpreted through Lawrence and Lorsch's (1967) distinction between differentiation and integration. Differentiation refers to the way organisational units develop different tasks, goals, time orientations, and ways of working (Lawrence & Lorsch, 1967). As differentiation increases, the need for integration also increases, because specialised units must still coordinate their contributions in relation to shared organisational tasks. In the studied organisation, increased functional specialisation has therefore created a stronger need to coordinate knowledge across mechanical engineering, electronics, software, production, testing, project management, and other related functions.

A central analytical point is that rapid growth changes the coordination requirements for knowledge integration. When the organisation was smaller, knowledge could more easily be accessed through proximity, direct interaction, and broader role understanding. Fewer employees and broader responsibilities meant that knowledge was closer to the situations in which it was needed. As the organisation has grown, knowledge has become more distributed across specialised roles and functions. This means that knowledge integration increasingly depends on mechanisms that actively connect knowledge across organisational boundaries.

This interpretation is also consistent with Greiner's (1972) view of organisational growth. Greiner (1972) argues that structures and practices that are effective in earlier phases of organisational development may become insufficient as the organisation grows. In the studied organisation, informal interaction and individual experience continue to support knowledge integration but the conditions under which these practices operate have changed. The organisation has become larger, more specialised, and more dependent on cross-functional coordination. As a result, earlier ways of integrating knowledge appear less sufficient.

Mintzberg's (1979) view of organisational structure further helps explain this pattern. Organisational structure concerns both how work is divided and how coordination is achieved between divided tasks (Mintzberg, 1979). In the studied organisation, rapid growth has increased the division of work, while the mechanisms for coordinating knowledge across this division appear to have developed more unevenly. This creates a tension between specialisation and integration. The organisation benefits from specialised expertise, but this expertise must still be connected to shared product and development tasks.

The analytical implication is that rapid growth increases the need for knowledge integration by making work more interdependent and knowledge more distributed. Growth does not reduce the importance of informal knowledge exchange, but it changes the conditions under which informal knowledge exchange takes place. When the organisation becomes more functionally divided, employees can no longer rely to the same extent on proximity, broad role understanding, or knowing who possesses relevant knowledge. Knowledge integration therefore requires more deliberate coordination mechanisms as organisational complexity increases.

5.2.2 Rapid growth fragments knowledge and capabilities

The second way in which rapid organisational growth affects knowledge integration is by fragmenting knowledge and capabilities. In the studied organisation, growth has contributed to a situation where knowledge is distributed across an increased number of employees, functions and organisational interfaces. This distribution in itself is not problematic. From a knowledge integration perspective, the problem arises when specialised knowledge becomes insufficiently connected across the organisation.

This can again be understood through Lawrence and Lorsch's (1967) argument that increased differentiation creates a corresponding need for integration. As functions become more specialised, they develop deeper knowledge within their own areas, but this also makes it more difficult to maintain shared understanding across the organisation. In the empirical findings, this was visible in how employees described a shift from working more closely around the product to working more separately within professional functions.

A central analytical point is therefore that rapid growth changes the structure of knowledge. Knowledge becomes less concentrated in shared product-related work and more distributed across functions, roles, and work packages. This makes knowledge integration more dependent on mechanisms that can connect specialised contributions across organisational boundaries.

Without such mechanisms, knowledge may still exist within organisations, but it remains fragmented in practice. This fragmentation also affects employees' understanding of the whole product and process. When employees work with narrower work packages, they may develop strong knowledge of their own task while losing insight into how their work connects to the broader development process. This weakens one of the conditions for coordination identified by Okhuysen and Bechky (2009) so called shared understanding. In this sense, growth affects knowledge integration by making it harder for employees to understand how their knowledge relates to the knowledge of others.

Carlile's (2004) perspective on knowledge boundaries also helps explain why fragmentation becomes problematic. When knowledge crosses functional or disciplinary boundaries, it is not enough to transfer information from one actor to another. Knowledge also needs to be translated and transformed so that it becomes meaningful across different functional perspectives (Carlile, 2004). In the studied organisation, increased functional separation means that more knowledge must cross such boundaries. If the organisation lacks mechanisms that support this translation, knowledge becomes located within functions rather than integrated across them.

The analytical implication is that rapid growth does not only increase the amount of knowledge that needs to be integrated, but it also changes how knowledge is organised. Knowledge becomes more specialised, more functionally located, and more dependent on cross-functional coordination. This makes fragmentation a central effect of rapid organisational growth on knowledge integration (Carlile, 2004; Okhuysen & Bechky, 2009).

5.2.3 Rapid growth exposes the limits of existing knowledge integration mechanisms

The third way in which rapid organisational growth affects knowledge integration is by exposing the limits of the organisation's existing knowledge integration mechanisms. As shown in Section 5.1 knowledge integration in the studied organisation is strongly supported by informal interaction, individual experience, personal networks, and partially formalised mechanisms. These mechanisms continue to be important, but rapid growth changes the conditions under which they operate.

From the theoretical framework, knowledge integration mechanisms can include both formal mechanisms, such as roles, routines, documentation, and process descriptions, and informal mechanisms, such as direct interaction, dialogue, and group problem-solving (Grant, 1996; Nonaka, 1994; Okhuysen & Bechky, 2009). In the studied organisation, both types of mechanisms are present, but the findings indicate that they do not fully align with the increased complexity created by rapid organisational growth.

Informal mechanisms remain important because much of the knowledge used in development work is tacit, situated, and experience-based (Lam, 2000; Nonaka, 1994). Direct interaction allows employees to clarify problems, identify relevant knowledge holders, and interpret knowledge in relation to specific work situations. However, as the organisation grows, these informal mechanisms become more difficult to rely on as the main route for knowledge integration. More employees need access to knowledge, functions become more specialised,

and knowledge is distributed across more organisational interfaces. This means that knowing whom to ask becomes less obvious, especially for new employees and employees outside established networks.

At the same time, formal mechanisms do not fully compensate for the limitations of informal integration. Documentation, internal webpages, process descriptions, templates, and onboarding practices could make knowledge more visible, accessible, and reusable across the organisation (Argote & Ingram, 2000; Grant, 1996). However, the empirical findings indicate that these mechanisms are not always sufficiently structured, updated, retrievable, or connected to actual work practices. This weakens their ability to support knowledge integration as the organisation grows.

Onboarding becomes particularly important in this context. Rapid growth increases the number of new employees who need to access organisational knowledge, understand work practices, and become able to contribute to shared development work. From a knowledge integration perspective, onboarding can support role clarity, social access, practical guidance, and knowledge sharing channels (Bauer et al., 2007; Fang et al., 2011; Klein et al., 2015). However, when onboarding is not sufficiently developed, newcomers become dependent on individual initiative, informal questioning, and the availability of experienced employees. This creates pressure on experienced employees. Their knowledge is needed both for ongoing development work and for supporting newcomers. As a result, knowledge integration becomes vulnerable when growth increases the number of employees who need knowledge faster than the organisation develops mechanisms for making that knowledge accessible. The issue is therefore not only that knowledge transfer becomes more difficult. The deeper issue is that knowledge internalisation becomes uneven, since employees may receive information without fully understanding how it should be applied in practice.

The analytical implication is that rapid growth exposes a mismatch between increasing organisational complexity and existing knowledge integration mechanisms. Informal mechanisms remain necessary, but they become less sufficient as the organisation expands. Formal mechanisms become more important, but they only support knowledge integration when they are embedded in actual work practices. In relation to RQ2, rapid organisational growth therefore affects knowledge integration by making the balance between informal and formal mechanisms more critical.

5.2.4 Summary of answer to RQ2

The analysis of RQ2 indicates that rapid organisational growth affects knowledge integration by increasing both the need for integration and the difficulty of achieving it. As the organisation grows, knowledge becomes more distributed across employees, roles, functions, projects, and organisational interfaces. This increases differentiation and creates stronger coordination demands.

Growth also contributes to fragmentation of knowledge and capabilities. Employees become more specialised and more focused on narrower work packages, while whole-product and

whole-process understanding becomes harder to maintain. This weakens shared understanding across functions and makes knowledge integration more dependent on deliberate coordination mechanisms.

Furthermore, rapid growth exposes the limits of existing knowledge integration mechanisms. Informal mechanisms remain important because they support situated judgement, tacit knowledge sharing, and practical problem-solving. However, they become less sufficient as the main route for knowledge integration when more employees need access to knowledge across a larger and more specialised organisation. At the same time, formal mechanisms become more necessary, but they only support knowledge integration when they are connected to actual work practices.

Overall, RQ2 is answered by identifying that rapid organisational growth creates a mismatch between increasing organisational complexity and existing knowledge integration mechanisms. Informal mechanisms remain important, but they become harder to rely on as the organisation expands. Formal mechanisms become more necessary, but they only support integration when they are practically usable and embedded in daily work. Rapid growth therefore affects knowledge integration by intensifying the tension between informal, person-dependent knowledge integration and the need for more visible, accessible, and scalable integration mechanisms.

5.3 Synthesis of Knowledge Integration During Rapid Organisational Growth

The analysis of RQ1 and RQ2 can be synthesised into a model of knowledge integration during rapid organisational growth. The model brings together the main empirical patterns from Chapter 4 and the theoretical interpretation developed in this chapter. It should therefore not be understood as an additional empirical finding, but as an analytical synthesis of how knowledge integration is characterised in the studied organisation and how these conditions are affected during rapid organisational growth.

The analysis shows that knowledge integration during rapid organisational growth is shaped by coexisting conditions. On the one hand, knowledge integration is largely individualised and informal. Employees rely on experience, personal networks, direct communication, and situated judgement to access, interpret, and apply knowledge in daily work. These practices support flexibility and problem-solving, especially when knowledge is tacit, context-dependent, or difficult to fully capture in documentation (Lam, 2000; Okhuysen & Bechky, 2009). Informal knowledge integration is therefore not a weakness. Rather, it is an important part of how specialised knowledge is coordinated and applied in innovation-driven work.

On the other hand, the organisation also uses formal knowledge mechanisms intended to make knowledge visible, accessible, and reusable. Documentation, internal webpages, process descriptions, templates, and onboarding routines are meant to support knowledge integration across individuals, roles, and functions (Argote & Ingram, 2000; Grant, 1996). However, the

analysis indicates that these mechanisms do not always correspond to how work is carried out in practice. When formal mechanisms are difficult to find, insufficiently updated, unclear, or disconnected from daily work, they provide only partial support for knowledge integration. As a result, employees continue to rely on informal and person-dependent ways of accessing and applying knowledge.

The central analytical concept in the model is friction. Friction refers to the tension that emerges when informal knowledge integration remains necessary, while formal knowledge mechanisms are not sufficiently aligned with work practice. This friction is not created by either informal or formal mechanisms alone. Rather, it emerges in the relationship between them. Informal knowledge integration enables employees to solve problems and adapt knowledge to specific situations, but it also makes knowledge access dependent on specific individuals, prior experience, and personal networks. Formal mechanisms could reduce this dependence, but only when they are practically useful and embedded in the way work is performed.

Rapid organisational growth intensifies this friction. As the organisation grows, knowledge becomes more distributed across employees, roles, functions, and projects. This increases the need for coordination, knowledge transfer, onboarding routines, and capability development (Bauer et al., 2007; Okhuysen & Bechky, 2009). At the same time, the informal networks and experience-based practices that previously supported knowledge integration become harder to maintain as the main route for accessing knowledge. More employees need access to knowledge, while experienced employees become increasingly important as knowledge holders, problem-solvers, and informal support for newcomers.

This means that rapid organisational growth does not simply create a need for more formalisation. Instead, it changes the balance between informal and formal knowledge integration. Informal mechanisms remain necessary because they support tacit knowledge sharing, situated judgement, and practical problem-solving. However, they become less sufficient when the organisation expands, and more employees need access to knowledge across a larger and more specialised organisational setting. Formal mechanisms become more important, but they only support knowledge integration when they are aligned with work practice and connected to how employees search for, interpret, and apply knowledge (Argote & Ingram, 2000; Grant, 1996; Okhuysen & Bechky, 2009).

The main analytical insight from this chapter is therefore that knowledge integration during rapid organisational growth is characterised by a tension between flexibility and scalability. Individualised and informal knowledge integration supports flexibility in daily work, while formal knowledge mechanisms are needed to make knowledge more accessible and reusable across a growing organisation. When these conditions coexist without sufficient alignment, friction emerges. During rapid growth, this friction contributes to fragmentation of knowledge and capabilities, as knowledge becomes increasingly distributed while the mechanisms for integration do not develop at the same pace. This synthesis is visualised in Figure 5.1 Model of knowledge integration during rapid growth.

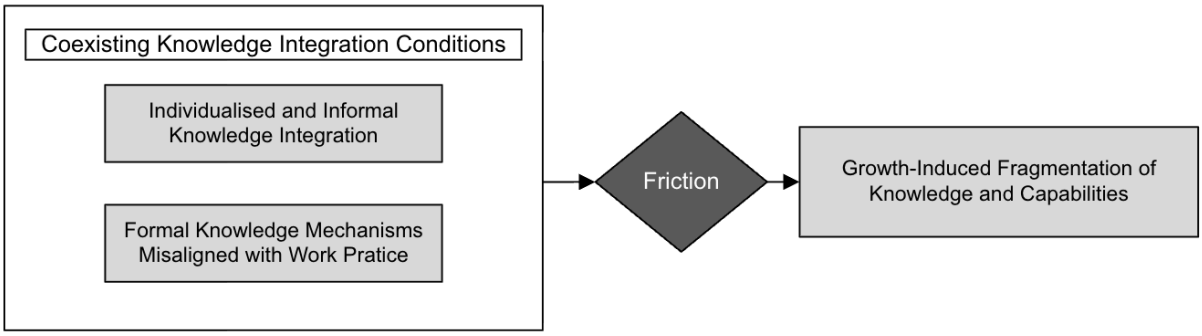


Figure 5.1 Model of knowledge integration during rapid growth

Comment. The model synthesises how coexisting conditions for knowledge integration create friction, which during rapid organisational growth contributes to growth-induced fragmentation of knowledge and capabilities.

6 Discussion and Conclusions

This chapter discusses the findings of the study in relation to its purpose. It presents the theoretical and practical implications, addresses the limitations of the study, suggests directions for further research, and concludes the thesis.

6.1 Discussion of Findings

The purpose of this study was to develop a better understanding of how innovation-driven organisations adapt knowledge integration during rapid organisational growth. The findings indicate that knowledge integration in the studied organisation is not absent. Employees integrate knowledge through experience, informal conversations, personal networks, and individual responsibility. These practices supported flexibility and problem-solving in daily work, particularly when knowledge was tacit, situated, and difficult to fully capture in documentation.

However, the same practices that enable knowledge integration also became vulnerable during rapid organisational growth. When knowledge was accessed primarily through personal networks and individual experience, employees needed to know who possessed relevant knowledge and how to reach them. This made knowledge integration effective for employees with established networks, but more difficult for newcomers or employees outside these networks. The issue was therefore not that informal knowledge integration was ineffective, but that it became uneven and person-dependent when it carried too much of the organisation's knowledge integration.

The empirical material also pointed to a gap between formal knowledge mechanisms and daily work practice. Documentation, internal webpages, process descriptions, templates, and onboarding routines were intended to make knowledge visible, accessible, and reusable. However, when these mechanisms were difficult to find, insufficiently updated, unclear, or disconnected from daily work, they provided only partial support. As a result, employees continued to rely on informal and experience-based practices to access, interpret, and apply knowledge.

Rapid organisational growth intensifies this situation. As the organisation grew, knowledge became more distributed across employees, roles, functions, and projects. This increased the need for coordination, knowledge transfer, onboarding, and capability development. At the same time, the informal networks and experience-based practices that previously supported knowledge integration became harder to maintain as the main route for accessing knowledge. This created a mismatch between increasing organisational complexity and existing knowledge integration mechanisms.

The main finding of the study is therefore that knowledge integration during rapid organisational growth is characterised by a tension between flexibility and scalability. Informal and individualised knowledge integration supports flexibility because it enables employees to interpret knowledge, ask questions, and solve situated problems. Formal mechanisms are

needed for scalability because they can make knowledge more visible, accessible, and reusable across a larger organisation. The challenge is not to replace informal knowledge integration with formal mechanisms, but to strengthen the interplay between them.

The interpretation shifts the discussion from whether knowledge integration should be informal or formal, to how informal and formal mechanisms can be connected during rapid organisational growth. Innovation-driven organisations need to preserve the informal interactions that make tacit and experience-based knowledge usable, while developing formal mechanisms that reduce person-dependence and support knowledge access across functions. This forms the basis for the theoretical and prescriptive implications discussed in the following sections.

6.2 Theoretical Implications

The study contributes to the literature on knowledge integration by illustrating how rapid organisational growth changes the conditions under which knowledge integration takes place. Previous research has described knowledge integration as the coordinated combination and application of specialised knowledge in collective work (Argote et al., 2022; Grant, 1996). This study supports this view but adds that the mechanisms used to integrate knowledge can become strained when organisational complexity increases faster than formal structures, routines, and onboarding practices develop.

A central theoretical implication is that informal knowledge integration should be understood as both enabling and vulnerable. Informal interaction, personal networks, and experience-based judgement are important for integrating tacit and situated knowledge in innovation-driven work. However, when informal mechanisms become the dominant way of locating and applying knowledge, they can create person-dependence and uneven knowledge access. This suggests that the effectiveness of informal knowledge integration depends on the organisational conditions in which it operates.

The study also contributes by highlighting the importance of alignment between formal mechanisms and work practice. Documentation, process descriptions, internal webpages, and onboarding routines do not support knowledge integration simply by existing. They become effective integration mechanisms when they are accessible, understandable, updated, and connected to how employees search for, interpret, combine, and apply knowledge in daily work. The theoretical implication is therefore that knowledge integration during rapid organisational growth depends on the interplay between informal and formal mechanisms, rather than on either type of mechanism alone.

6.3 Practical Implication: Practice-Aligned Formalisation

The practical implication of this study is practice-aligned formalisation. This refers to the need for innovation-driven organisations during rapid organisational growth to strengthen formal

knowledge mechanisms in ways that support, rather than replace, the informal and situated knowledge integration already taking place in daily work.

The findings indicate that informal interaction, personal networks, and experience-based judgement remain important because much of the knowledge used in innovation-driven work is tacit, situated, and difficult to fully codify. Employees need to ask questions, interpret previous experience, discuss practical problems, and translate knowledge across roles and functions. However, the findings also indicate that informal knowledge integration becomes more vulnerable as the organisation grows. When more employees, roles, functions, and projects become involved, knowledge becomes more distributed and harder to locate. Relying heavily on informal networks can therefore create person-dependence, uneven access to knowledge, and difficulties for newcomers in understanding where relevant knowledge is located.

Practice-aligned formalisation therefore implies that formal mechanisms should be developed from, and aligned with, actual work practices. Documentation, process descriptions, onboarding, and cross-functional forums should support how employees search for, interpret, combine, and apply knowledge in practice. This adaptation is visualised in Figure 6.1.

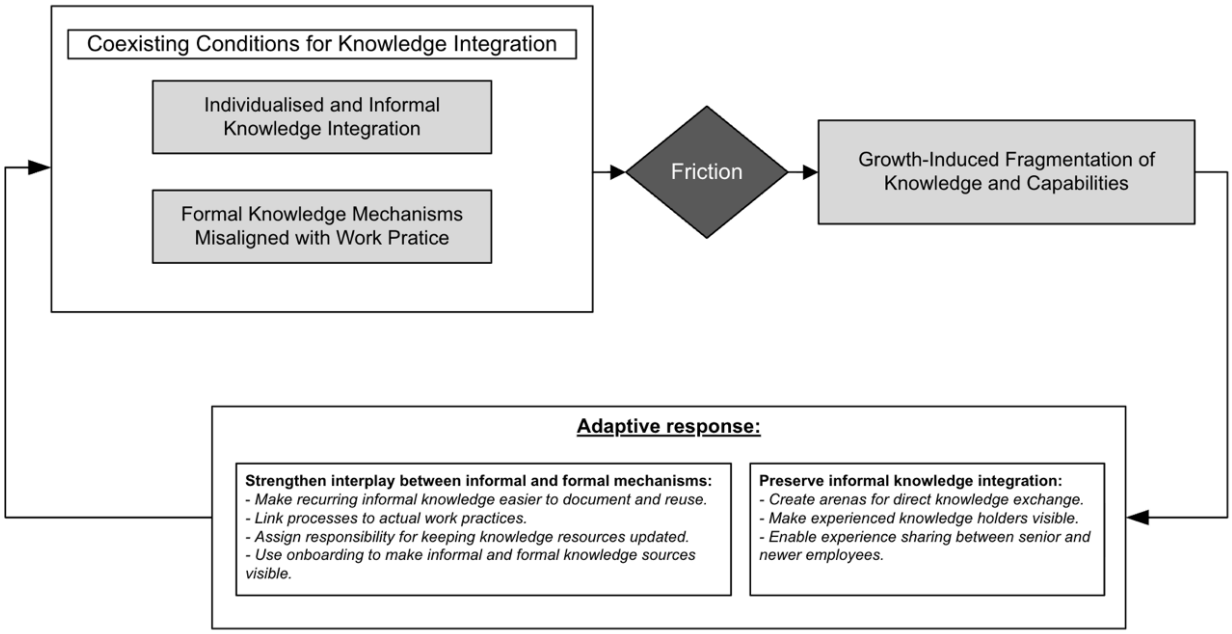


Figure 6.1 Adaptation of knowledge integration during rapid growth

Comment. The model illustrates how coexisting conditions for knowledge integration can create friction during rapid organisational growth, contributing to growth-induced fragmentation of knowledge and capabilities. The adaptive response is practice-aligned formalisation, where informal knowledge integration is preserved while formal mechanisms are strengthened to make knowledge more visible, accessible, and reusable in daily work.

In practice, this has several implications. First, documentation should be treated as a working resource rather than as an archive. Reports, templates, internal webpages, and other knowledge resources should help employees understand where relevant knowledge is located, how it has been used before, and how it can guide current work. For documentation to support knowledge integration, it needs to be accessible, understandable, updated, and connected to actual work practices. This also requires clear responsibility for keeping knowledge resources updated and relevant to current work practices.

Second, process descriptions should support coordination rather than only describe formal steps. In rapidly growing organisations, high-level process descriptions may be insufficient if employees still need to interpret what should be done, who should be involved, and where relevant knowledge can be found. Process descriptions should therefore clarify responsibilities, expected inputs and outputs, decision points, and relevant knowledge sources. This is especially important when growth increases specialisation and employees become more dependent on knowledge held by others.

Third, onboarding should be treated as a knowledge integration mechanism. New employees need to understand not only their formal role, but also where knowledge is located, who holds important experience, how different roles and functions interact, and how informal and formal knowledge sources can be used in practice. This also means making experienced knowledge holders more visible and creating opportunities for experience sharing between senior and newer employees. In this way, onboarding can reduce dependence on individual knowledge holders and support more equal access to organisational knowledge.

Finally, cross-functional forums should not simply create more meetings. Their purpose should be to support shared understanding, clarify dependencies, discuss lessons learned, and translate knowledge across roles and functions. Such forums can help preserve the flexibility of informal knowledge exchange while making important knowledge less dependent on individual networks.

The model should not be understood as an empirically tested solution. Rather, it is a practical and prescriptive implication derived from the analysis. It suggests that innovation-driven organisations during rapid organisational growth can adapt knowledge integration by strengthening the interplay between informal knowledge practices and formal knowledge mechanisms, which may also be seen as indirectly supporting organisational conditions for sustainable industrial innovation in line with SDG 9 (United Nations, n.d.).

6.4 Limitations and Further Research

The study has limitations that should be considered when interpreting the findings. First, the study was conducted as a qualitative single-case study within one innovation-driven organisation. This enabled a deeper understanding of knowledge integration in its organisational context, but it also limits the transferability of the findings. The findings should

therefore not be understood as statistically generalisable but as analytically generalisable to theory and potentially transferable to similar organisational contexts.

Second, the empirical focus was limited to mechanical development and design-test-related work. This provided a focused setting for examining how knowledge is accessed, shared, coordinated, and applied in practice. However, knowledge integration may be experienced differently in other functions, such as software, electronics, HR, or management. The absence of these perspectives means that the findings mainly reflect knowledge integration from the perspective of mechanical development.

Finally, the proposed model of practice-aligned formalisation is a practical and prescriptive implication derived from the analysis. However, it should not be understood as an empirically tested solution. Further research could address these limitations through multiple-case studies in comparable innovation-driven organisational contexts during rapid organisational growth. Following a replication logic, such studies could examine whether the tension identified in this study are replicated in similar settings. Longitudinal case studies could also examine how practice-aligned formalisation develops over time and whether it can reduce person-dependence while preserving informal knowledge integration needed for innovation-driven work.

6.5 Conclusions

The purpose of this study was to develop a better understanding of how innovation-driven organisations adapt knowledge integration during rapid organisational growth. In relation to RQ1, the study concludes that knowledge integration was characterised by informal and experience-based coordination across roles and functions. Knowledge was often accessed and applied through interaction with experienced colleagues rather than through formalised knowledge sources alone. This made knowledge integration flexible and closely connected to practical problem-solving. However, it also made knowledge integration dependent on individuals, personal networks, and employees' ability to know where relevant knowledge could be found.

In relation to RQ2, the study concludes that rapid organisational growth made knowledge integration more difficult by increasing the distribution of knowledge across employees, roles, functions, and projects. As the organisation grew, existing informal mechanisms became more vulnerable because knowledge became harder to locate, coordinate, and reuse. Growth also increased the need for formal mechanisms, but these mechanisms did not automatically support knowledge integration when they were disconnected from actual work practices.

The overall conclusion is that innovation-driven organisations during rapid organisational growth need to adapt knowledge integration by strengthening the interplay between informal knowledge practices and formal knowledge mechanisms. This does not mean replacing informal interaction with formalisation. Rather, formal mechanisms need to be aligned with daily work practices so that knowledge can become more visible, accessible, reusable, and connected to how employees search for, interpret, combine, and apply knowledge in practice.

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Appendices

Appendix A – Interview guide

Appendix B – Information letter and consent form

Appendix C – Data structure and supporting empirical evidence

Appendix A

Introductory question	**Q1:** How long have you worked in the organisation?
Part 1: Role and work context	**Q2:** Can you describe your role in the organisation? **Q3:** In what way are you involved in work that contributes to developing or defining a design test? **Q4:** Which part of this work are you usually most involved in? • Before, during, or after the design test?
Part 2: Understanding the concept of design test	**Q5:** What does a design test mean in your work? **Q6:** What does a design test usually include? What type of information or decisions are involved? **Q7:** How is it decided which design tests are relevant? • How is a user profile identified when selecting relevant design tests?
Part 3: How the work is carried out	**Q8:** Can you describe an example of when you worked on developing or defining a design test? **Q9:** How did the work begin in that case? **Q10:** How did the work develop until the design test was defined? • At what point in this work do you need to bring together knowledge or information from several people or functions?

Part 4: Actors and roles	**Q11:** Which roles are usually involved in this work? **Q12:** Which roles do you most often collaborate with in this work? **Q13:** If uncertainties arise, which roles do you contact?
Part 5: Knowledge in the work	**Q14:** What type of knowledge or experience is needed to carry out this work? **Q15:** How do different people or areas of expertise contribute to the work? **Q16:** Can you give an example of a situation where people with different areas of expertise needed to collaborate in order to define a design test? • How would you describe the way knowledge from different people is brought together in practice?
Part 6: Communication and coordination	**Q17:** How do different roles and/or functions collaborate and communicate in this work? • Which departments are usually involved in this work? **Q18:** How do you know who to turn to when you need specific knowledge or experience? • How is knowledge from this work shared with other parts of the organisation? • How does this collaboration work in practice? • What makes it easy or difficult to bring together information or knowledge from different departments?

Part 7: Challenges	**Q19:** What challenges can arise in this work? • Can you give a concrete example? • Are these challenges related to how you collaborate or share information/knowledge? • When do these challenges usually arise?
Part 8: Change over time	**Q20:** How has this work changed over time since you started? **Q21:** How has organisational growth affected how you collaborate and share knowledge? • What is the biggest difference today compared to before when it comes to collaboration and information sharing?
Part 9: Improvement	**Q22:** What works well today when it comes to combining knowledge between different people and departments? **Q23:** What would need to be improved? **Q24:** If the organisation continues to grow, what needs to change in how you collaborate and share information/knowledge?
Part 10: Identifying additional interview respondents	**Q25:** Are there any other people you think I should talk to?

Appendix B

Information Letter and Consent Form

Background and purpose of the study

In innovation-driven organisations, knowledge plays a crucial role in the development of new products, technologies and solutions. When organisations grow rapidly, knowledge is often spread across different individuals, teams and functions.

To carry out complex development activities, organisations must therefore combine knowledge from multiple sources. In the field of knowledge management, this process is called knowledge integration, which means that knowledge from different individuals and areas of expertise is brought together to solve a specific task. This study focuses on a specific organisational process in which real-life training scenarios are translated into defined design tests for a product that aims to simulate these scenarios.

The purpose of the study is to investigate how knowledge from different individuals and areas of expertise is combined and integrated in this process, particularly in the context of a rapidly growing innovation-driven organisation. By interviewing people who are involved in or familiar with this process, the study aims to develop a better understanding of how knowledge is gathered in practice when complex development tasks are performed.

How the study is performed

To fulfil the research purpose, data will be collected through semi-structured interviews, document studies, and observations. Participants will take part in interviews lasting approximately 30-45 minutes. The interviews can be conducted face-to-face or digitally, depending on availability and preference.

During the interview, participants will be asked to share their experience, perspectives, and how knowledge from different individuals or areas of expertise is combined in this work.

With the participants' permission, the interviews will be recorded to ensure transcription and analysis. Participation in the interview requires no preparation, and participants are encouraged to speak freely about their experiences and opinions.

Management of data and privacy

The collected data will be handled in accordance with the EU General Data Protection Regulation (GDPR). The information gathered during the interviews will be treated confidentially and stored securely. Participants will not be identified by name in the interview transcript or in the final thesis. Instead, participants will be assigned a code, and the list linking names to codes will be stored separately.

The interviews will be recorded with participants consent to ensure accurate transcription. The recording will only be used for transcription and will be deleted after the study is completed.

The study is expected to be completed in June 2026. The research results will be presented in a master's thesis at Jönköping University, which may also be shared with relevant stakeholders within the organisation.

Voluntary participation

Participation in this study is voluntary. Participants have the right to decline participation or withdraw from the study at any time without giving any reason. Choosing not to participate or withdrawing from the study will not have any negative consequences and will not affect the participants relationship with the researcher or the organisation.

Responsible for the study

This study is conducted on behalf of Jönköping University by the following researcher:

Alexandra Fredriksson

Tel: -

Email: -

If you have any questions about the study, you are welcome to contact me.

Consent form

I have received written and oral information about the study and have been given the opportunity to ask questions. I agree to participate in this study on how knowledge from different individuals and areas of expertise is combined in a specific development process within the company.

I consent to the processing of my personal data as described in the information letter.

Date: **Signature:**

_____ _____

Appendix C

Aggregate dimension: Individualised and informal knowledge integration	
Second-order theme	Selected evidence of first-order concepts
Experience-based knowledge integration	**Carrying experience into new projects**
	1. "Partly, you know what holds up and what breaks, that's experience you carry with you, so you also know, more or less, what kinds of tests you need to run." (Interview: Respondent A) 2. "You have to be a bit clever, or at least avoid starting everything from scratch again, because you also bring some experience with you." (Interview: Respondent A) 3. "Experience is something we need to carry with us somehow and bring into a project." (Interview: Respondent C) 4. "The process exists within individuals, and now it's difficult as we're getting a lot of new people." (Observation 2: Workshop) 5. "The process of ... existed within individuals, not clearly documented." (Observation 1: Task force meeting)
	Relying on prior experience when making decisions
	6. "But you realise those things afterwards, really. Partly, you rely on your experience, and partly, you think things through when you have the materials in front of you what should I do? What do I need to test to feel confident here?" (Interview: Respondent A) 7. "Or you simply have to guess? Well, not guess exactly, but you draw on your previous experience." (Interview: Respondent A) 8. "With experience you also know a bit that this particular test is really nothing out of the ordinary." (Interview: Respondent A)

	9. “A more senior engineer might have a better understanding of how the product is used in the field and what requirements are placed on it. They already know that, without necessarily having to read the requirement specification in detail.” (Interview: Respondent F)
	Understanding task requirements through experience
	10. “This is the kind of experience you’d say they need in order to be able to design.” (Interview: Respondent C) 11. “For those who write the specifications, they need to have that understanding in order to write a good spec.” (Interview: Respondent D) 12. “I’ve used it myself, that's how it’s been, because there haven't been that many before.” (Interview: Respondent A) 13. “If you have that kind of experience, if you’ve been a design engineer yourself and been involved in many development projects then you can identify shortcomings, and sometimes you can step in and steer things...” (Interview: Respondent G) 14. “Many new designers might not have the experience ... I think it’s really useful to consult ... early on.” (Interview: Respondent E)
Second-order theme	Selected evidence of first-order concepts
Informal network-based knowledge integration	**Asking colleagues informally when solving problems**
	15. “So that part being able to ask, ‘What did you do here?’ and so on is really useful. These aren’t official meetings, but rather informal chats where you come down to talk, and maybe you’ve run into a problem, and then someone can say, ‘Well, you could do it like this’.” (Interview: Respondent B) 16. “You just go to colleagues sitting nearby who you thought might be able to help.” (Interview: Respondent D) 17. “Depending on what kind of construction it is or what problems you see, and it could be with.... or it could be with ...you discuss it. How should this work? How could this work?” (Interview: Respondent A)

	Seeking specific individuals for help
	18. “Personal networks and it’s from the site. They call me directly because they know me, and then they get my answer.” (Interview: Respondent C) 19. “Then you can write as many documents and processes as you like, but if I need someone to do something on the electronics side, I already know who to go to.” (Interview: Respondent E) 20. “Depending on what kind of construction it is or what problems you see, and it could be with xxx or it could be with yyy, you discuss it.” (Interview: Respondent A) 21. “If we have problems with the mechanics, the first thing you check is, of course, the drawings to make sure they’re correct. But if there’s still a problem, then you have to go and talk to the designer, and together you decide how to solve it in the short term and how to address it in the long term.” (Interview: B) 22. “To those of us sitting here, I'm new and don’t really know who’s good at what. Who among you are the experienced ones? Who of you has the knowledge?” (Observation 2: Workshop)
Second-order theme	Selected evidence of first-order concepts
Individualised knowledge integration under time pressure	**Taking individual responsibility to find knowledge**
	23. “Depending on who you ask, you get different answers.” (Interview: Respondent H) 24. “But in practice, it’s more your own responsibility to reach out, ask, and get their input.” (Interview: Respondent H) 25. “You tend to ask the people closest to you. But yes, in that sense, I’ve gone to different people at times and received different answers, so there hasn’t really been any specific approach I’ve followed.” (Interview: Respondent H)

	26. “No, but then I might ask a senior engineer about something say, an adhesive, and they’ll tell me to go and speak to another senior engineer who can help me with it.” (Interview: Respondent D) 27. “I know that someone has the knowledge about this, and I can ask that person, but is there a formal forum for this?” (Observation 2: Workshop) 28. “Lately, it’s felt like there’s been a lot of new things coming in, and a lot of new people as well, and that they’ve mostly just had to jump in, learn on their own, and take individual responsibility.” (Interview. Respondent H)
	Prioritising delivery over knowledge exchange
	29. “Something we maybe didn’t have, since we weren’t that many before ... was proper knowledge sharing and exchange. And we also didn’t have the time for it, because we had to focus on projects all the time.” (Interview: Respondent F) 30. “Many new employees at the company, difficult to transfer knowledge” (Observation 1: Task force meeting) 31. “Because that’s how it is, we don’t have all the time in the world in the project ... the possibility to do really good design tests just isn’t there anymore. There isn’t enough time. Yes, because by the time you have the things you’re supposed to test, it’s already time to start building them.” (Interview: Respondent A) 32. “We’ve effectively had 110% workload. We’ve had customers to deal with… we’ve been travelling every other week for quite a long time, and we simply haven’t had the time to support development, neither in terms of knowledge transfer nor testing to really support it. We just haven’t had the time for that, and it hasn’t been our choice.” (Interview: Respondent C) 33. “Lately, I’ve felt that, as I said, the projects are being rushed, and you don’t really get the time to properly verify steps in

	process. Sometimes it's so stressful that you don't have time at all ... And because of that, you don't really have time to run those kinds of tests either, for that matter." (Interview: Respondent H) 34. "According to our processes, we're supposed to have quite a few ... phases where we verify...But a lot of that doesn't really get done due to time constraints. That's probably the part that ends up being shortened, since testing takes time." (Interview: Respondent H)
	Coping with high workload and constant time pressure
	35. "And then we end up in that situation time and time again suddenly everything becomes urgent: 'we need to deliver tomorrow,' and we haven't .... yet. And at the same time, we're supposed to carry out tests that take two to three months in calendar time to complete. That equation is often very difficult to make work." (Interview: Respondent E) 36. "We were quite heavily pressed in terms of time because we simply had too much to do." (Interview: Respondent F) 37. "We find ourselves in a situation where we have grown quite rapidly, with more projects, more staff and a faster pace." (Observation 2: Workshop) 38. "And ultimately, it's about freeing up time for what really adds value. Because if the process doesn't help us, we won't use it." (Observation 2: Workshop) 39. "We have been under considerable time pressure as orders have come in fast and as a result, our focus on innovation has slipped, as we haven't had the time to concentrate on innovative development." (Informal interview) 40. "Competencies need to be available at the same time, so they can sit together and develop things." (Interview: Respondent C)
Aggregate dimension: Formal knowledge mechanisms misaligned with work practice	

Second-order theme	Selected evidence of first-order concepts
Weak knowledge codification and retrieval	**Lacking structure for documenting knowledge**
	41. “I haven’t found any general structure for how such a report should be written; it’s more based on your own judgement, I think.” (Interview: Respondent H) 42. “I would be very surprised if anyone goes in and reads old reports, it’s probably quite heavy going.” (Interview: Respondent B) 43. “There’s no real issue getting this written down, and that’s where I feel I could have a role, as someone who is good at writing documentation.” (Interview: Respondent C) 44. “Some pages include information about the last update, whilst the majority of pages do not have a date for last update.” (Document study: Internal website) 45. “There is some documentation, but it is not clearly defined.” (Observation 2: Workshop) 46. “Comment on internal website: Is it clear that everyone has responsibility to update this website with information and knowledge?” (Informal interviews)
	Lacking guidance for locating knowledge
	47. “It’s doubtful whether there’s any documentation for that…” (Interview: Respondent D) 48. “So, it’s stored in our system, but yes, I believe that all stakeholders should have access. I can’t say for sure. I’m not a hundred percent certain but I think so.” (Interview: Respondent F) 49. “Where is the documentation stored?” (Observation 2: Workshop) 50. “Where do I find templates for this?” (Observation 2: Workshop)
	Assessing documentation as inconsistent or incomplete

	51. “Some pages include information about the last update, whilst the majority of pages do not have a date for last update.” (Document study: Internal website) 52. “One webpage has been marked in red text as ‘out of date, needs to be updated’.” (Document study: Internal website) 53. “Some webpages contain no more than a single sentence.” (Document study: Internal website) 54. “It is a cluster of links, and there is no clear structure for how you find things and what is actually available.” (Document study: Internal website) 55. “Some links lead nowhere … no clear structure.” (Document study: Internal website)
Second-order theme	Selected evidence of first-order concepts
Decoupled formal processes and situated knowledge practices	**Making sense of process requirements**
	56. “If you look at the process and click on it, what do you actually get? Do you get templates, or what is it that comes up? You can see that the phase exists, but not really what should be done.” (Interview: Respondent G) 57. “Lack of clarity regarding steps, responsibilities and expectations.” (Observation 2: Workshop) 58. “A year ago, we sat down and tried to map things out using the process, but we couldn't get it to work.” (Observation 2: Workshop) 59. “Same terms for different things are used.” (Document study: Mechanical Process) 60. “Variation in content available under each phase.” (Document study: Mechanical Process) 61. “Various interpretations of terms within the process are brought up and questioned.” (Observation 2: Workshop) 62. “Different interpretations of terms within the process are highlighted...confirming that concepts relating to testing vary.” (Observation 1: Task force meeting)

	63. “Multiple questions are asked about the process.” (Observation 1: Task force meeting) 64. “Broad categories as input and output.” (Document study: Mechanical Process) 65. “Some phases can be clicked and further described.” (Document study: Mechanical Process) 66. “Some phases are not clickable and are not further described, just a headline describing the phase.” (Document study: Mechanical Process) 67. “Some parts are described using abbreviations or a descriptive word followed by an abbreviation.” (Document study: Mechanical Process) 68. “The process is confusing and is not easy to understand.” (Observation 3: Workshop) 69. “Why are there not explanations for each phase and what is supposed to be done? And why is there a lot of abbreviations? Is there any document or something to explain these abbreviations?” (Observation 3: Workshop)
	Working with poorly embedded processes
	70. “I’ve never been part of any project so far that has followed the process completely.” (Interview: Respondent H) 71. “Process was created and formed without inclusion of us.” (Observation 2: Workshop) 72. “We haven’t been working according to this process.” (Observation 2: Workshop) 73. “In practice, it’s not as easy as it seems from looking at the process.” (Observation 2: Workshop) 74. “This process has never been properly introduced to the employees; I have not had a formal introduction to the process.” (Observation 2: Workshop)
	Solving work through ad-hoc collaboration

	75. “No, it’s… in some projects we’re involved, and in others we’re not, so… yes, that’s actually a good question that I don’t really have full insight into.” (Interview: Respondent F) 76. “Well, then there are these design reviews ... with input from production, after-sales, as well as ... and even the .....” (Interview: Respondent D) 77. “Sometimes you hand it over to...the environmental testing department, and ask, could you test this for me?” (Interview: Respondent A) 78. “The interface, who I work with varies. These aren’t official meetings either, it’s simply about dealing with whatever comes up. It can be anything from stores, to purchasing, to ..., to details that need fixing on..., design, electronics, towards production engineering, towards platform, towards application so we have a lot of contact points”. (Interview: Respondent B) 79. “Establish a consistent approach and ensure clarity in projects.” (Observation 1: Task force meeting) 80. “There is a need to keep a continuous communication with people from different units to provide a solid foundation for formal meetings.” (Observation 2: Workshop) 81. “There are several different roles or rather competencies needed to develop a finished product.” (Interview: Respondent E)
Aggregate dimension: Growth-induced fragmentation of knowledge and capabilities	
Second-order theme	Selected evidence of first-order concepts
Fragmentation of work and knowledge across functions	**Working in separate functions with limited interaction**
	82. “Now you have mechanical engineers on one floor, doing their own thing, living in their own world. Then you have the electronics people doing the same, and software doing their own thing as well. We’re not working together around the product. We’re organising ourselves around our professions instead. And then the project managers are sitting somewhere else…” (Interview: Respondent C)

	83. “Product teams sat together as a group around one product, not divided by field.” (Interview: Respondent C) 84. “Now people sit in their own areas someone on the systems side working on something in isolation and then sending it out…” (Interview: Respondent E) 85. “Yes, it’s much more spread out now, and I don’t think we coordinate the available competencies as well as we used to. Whether that’s because we’ve become so many more people, it’s hard to say. But I think that’s something all companies face when they grow very quickly, it doesn’t happen without problems, and not everyone will like or be comfortable with how things turn out.” (Interview: Respondent E) 86. “It used to be a completely different set-up; now things are a bit more divided. In the past, we were a homogeneous group... all in the same place.” (Interview: Respondent B) 87. “This creates silos anyway, because we are not informed or included when the new concept is started and when the information reaches us, there is more of a demand that they need resource from us, and our resources do already have a high workload.” (Informal interview: Cross-functional work) 88. “There weren’t that many of us back then, and everyone basically did everything. Now we’re more divided, so organisationally, we work differently.” (Interview: Respondent A)
	Losing the overall understanding of product and work
	89. “In the project, we get work packages instead. Previously, we were more involved in the project as a whole, so to speak. Now, it’s more about work packages being handed out like this, and as a result, you lose the overall picture as well, since you’re not as involved in... in the entire process. Instead, you’re only given a small part, and you lose sight of what’s important well, you simply don’t get the full picture.” (Interview: Respondent A)

	90. “We get involved in the xxx series, and sometimes we get to take a look at an earlier stage and see a concept, but we don’t know when they’re developing things.” (Interview: Respondent B) 91. “Not everyone in the organisation is familiar with it.” (Observation 2: Workshop) 92. “As a result, the working methods we use can vary from person to person, which leads to confusion and sometimes unnecessary work.” (Observation 2: Workshop) 93. “Used to have coffee breaks, that is, informal meetings, all the time, and they were really useful.” (Interview: Respondent B) 94. “We solved a lot of things over coffee.” (Interview: Respondent B)
Second-order theme	Selected evidence of first-order concepts
Breakdown in knowledge transfer and internalisation	**Struggling to transfer and retain knowledge**
	95. “Many new employees at the company, difficult to transfer knowledge.” (Observation 1: Task force meeting) 96. “The company has grown and in the growth process, its close... and understanding of how the product is used have been lost.” (Observation 1. Task force meeting) 97. “All of it and the idea that there should be someone who has it all documented. Of course, we need documentation but trying to document a process like that the kind of knowledge we’re talking about, we’ve been trying to do that for as long as I’ve been at the company and it still hasn’t really worked.” (Interview: Respondent E) 98. “Yes, it was probably about 15 years since we last developed a xxx before that one. Something like that. So, it’s also something to reflect on, if you only develop something occasionally, how do you maintain that competence over time? People retire and others leave… so it becomes quite an important issue for the company.” (Interview: Respondent E)
	Lacking structured onboarding and training

	99. “Neglected newcomers too many new hires, not enough training.” (Interview: Respondent C) 100. “No, but I have often said that new employees must receive a proper introduction.” (Interview: Respondent A) 101. “It’s really important that new employees get out and have the opportunity to see how things are used it’s really important. It’s also crucial to see how things are assembled or repaired. You need that knowledge; you have to broaden your understanding in order to do your job well.” (Interview: Respondent A) 102. “But they should really be part of a team, a team that helps with drawing reviews, supports test planning, reviews test results, and so on ...you don’t benefit from all the experience that exists across the department.” (Interview: Respondent G) 103. “When someone joins, whether they’re a newly graduated engineer or someone with 20 years of experience in something else, there should be some kind of plan in place even before their first working day.” (Interview: Respondent E) 104. “Back then, we had an introduction programme where you were supposed to visit, for example, production for a week and do different tasks, and then you would fill in and tick off boxes to say, 'I’ve done this,' and after that, the introduction was complete.” (Interview: Respondent E) 105. “So, the easiest option for him is to ask me about everything, whenever he comes across something he doesn’t know how to handle. And that puts quite a bit of pressure on me. So, it’s really this onboarding phase that’s a bit challenging new people may find it difficult to know where to find things.” (Interview: Respondent H) 106. “Lately, it’s felt like there’s been a lot of new things coming in, and a lot of new people as well, and that they’ve mostly just had to jump in, learn on their own, and take individual responsibility.” (Interview: Respondent H)

	107. "And when I was new as well, I wasn't really aware that there were these useful pages you could rely on. So, you would usually just ask a more senior colleague how to do something. Either they would show you directly, or they would point you to where the guide is, and then you realise that you could have found out how to do it yourself there. But it's really about being able to find those things and getting a better introduction in some way." (Interview: Respondent H) 108. "Lack proper embedding and introduction and training within the department." (Observation 2: Workshop) 109. "Many new people here now and not many left with proper training. That makes things a bit difficult, as it can be hard to consult someone who doesn't really have the knowledge." (Interview: Respondent B) 110. "Training packages to train all involved stakeholders." (Observation 1: Task force meeting) 111. "We want to establish a way of working that is clear, that we all understand, and that actually helps us in our day-to-day work." (Observation 2: Workshop)